









# Modelling Financial Markets

## Chapter 18: Systemic Risk and Networks



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# When Does One Bank's Distress Become the System's Problem?

## From one bank to the system

- **Systemic risk**: the risk that the distress of one or several institutions impairs the financial system as a whole
  - and, through it, the real economy
- An **externality**: a bank does not pay for the losses its failure imposes on others
- Three families of measures (survey: Benoit, Colliard, Hurlin & Pérignon (2017))
  - **Exposure**: what does the bank lose when the market crashes?
  - MES: its average loss on crash days; SRISK: its capital shortfall in a crisis
  - **Contribution**: how much worse is the system when the bank is in distress?
  - $\Delta$CoVaR, with CoVaR the Value-at-Risk (VaR) of the system conditional on the bank
  - **Connectedness**: through which links do shocks travel?
  - Spillover and tail networks

# The Plan of the Chapter: Tail Losses and Financial Networks

## Two parts

- Part I: from the data to bank-level measures of systemic risk
  - data and conventions
  - exposure: MES, LRMES (the MES of a crisis lasting months), SRISK
  - contribution: CoVaR, $\Delta$CoVaR, and how far rankings can be trusted
- Part II: networks, stress tests and a research case
  - spillover networks and the Financial Risk Meter
  - stress tests and climate risk
  - a case study on quantile connectedness
  - one pass of the discovery loop with AI (artificial intelligence)
- Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

# Bank Runs: Northern Rock 2007 and Silicon Valley Bank 2023




Depositors queue outside a Northern Rock branch, Brighton, 14 September 2007

Photo: Dominic Alves (2007); CC BY 2.0; Wikimedia Commons

- Northern Rock (2007): a British mortgage lender
  - hit by the freeze in funding markets after the US subprime losses
  - depositors queued to withdraw their savings
- SVB (2023): a digital run
  - the FDIC (Federal Deposit Insurance Corporation) took the bank over on 10 March FDIC (2023)
  - two days later all depositors of SVB and Signature Bank were protected on **systemic** risk grounds Federal Reserve Board (2023)

## Through Which Channels Does Contagion Travel?

- One bank's distress becomes systemic through three channels
  - **Funding runs**: depositors and lenders withdraw from any bank that looks similar
  - **Common exposures** and **fire sales**: forced selling lowers prices for every holder
  - **Counterparty links**: one bank's liability is another bank's asset, so a default passes losses to its creditors

# Thirteen Banks, Three Markets: Prices and Log Returns

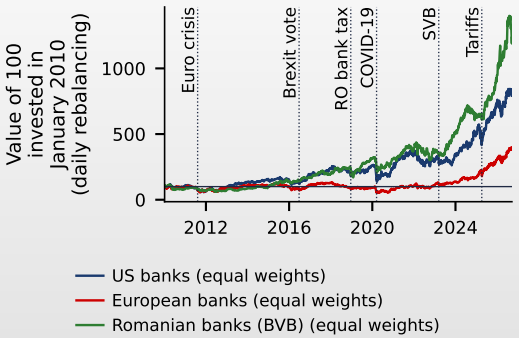
| Group | Banks (ticker, used in some charts) | Market index |
|---|---|---|
| US banks | JPMorgan (JPM), Bank of America (BAC), Citigroup (C), Goldman Sachs (GS), Morgan Stanley (MS), Wells Fargo (WFC) | S&P 500 |
| European banks | Deutsche Bank (DBK), BNP Paribas (BNP), Santander (SAN), ING (INGA), HSBC (HSBA) | Euro Stoxx 50 |
| Romanian banks | Banca Transilvania (TLV), BRD-Groupe Société Générale (BRD; "BRD-Groupe SG" in charts) | BET |
| Volatility | the VIX index (implied volatility of the S&P 500) | |

- Daily prices, January 2010 to 18 September 2026, adjusted for dividends and splits
  - Aligned on 3 960 common trading days first, then daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %
- Special cases
  - Romanian banks, traded on the Bucharest Stock Exchange (BVB, index BET): days without trades removed
  - Historical stress scenarios for the international banks start in 2007

## Losses as Positive Numbers and the System Portfolio

- Return $X$, loss $L = -X$
  - the level $\alpha$ is a **tail probability** (Chapter 7)
  - $\mathrm{VaR}_\alpha(X) = -q_\alpha(X)$, with $q_\alpha$ the $\alpha$-quantile of $X$
  - We write VaR 1%, CoVaR 1%, MES 5%, all positive when they are losses
- **System** return $X^{sys}$: the equal-weighted portfolio of the other banks of the same region
  - $X^{sys} = \frac{1}{n}\sum_j X_j$, a first-order approximation of the log return of the daily rebalanced portfolio
  - CoVaR is stated for this aggregate, while MES conditions on the regional market index

# Three Banking Systems, 2010–2026




- Equal-weighted portfolios of the banks of each region, rebalanced daily
  - Vertical lines mark the episodes used in this chapter
- 100 invested in January 2010 became 791 (US), 384 (Europe), 1 233 (Romania)
  - Maximum drawdown: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
  - Annualised volatility: 29%, 30%, 23% (market indices: 18%, 21%, 18%)
- The falls are **simultaneous**: euro crisis, COVID-19, 2023 bank failures
  - Romanian banks also have their own shock: the bank-tax announcement of December 2018 (label "RO bank tax")

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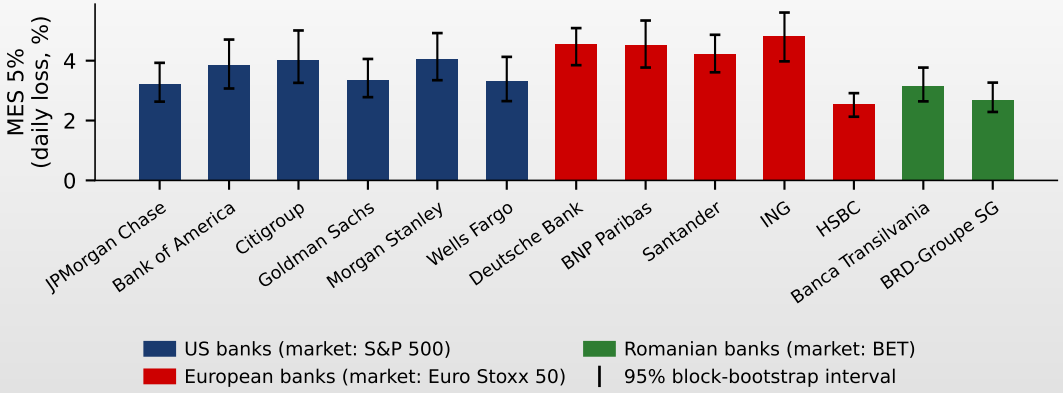
## How Much Does a Bank Lose When the Market Crashes?

- **Exposure**: the loss of the bank given a market crisis, not its stand-alone VaR
  - What matters are the banks that run out of capital exactly when the system needs it
- Three steps, from one crash day to a crisis
  - MES: the average loss of the bank on the worst 5% of market days, the level used by Acharya, Pedersen, Philippon & Richardson (2017)
  - LRMES: the loss of the bank in a crisis lasting weeks or months
  - SRISK: the capital the bank would lack after that loss

# Marginal Expected Shortfall and the Euler Property

- **MES** (Marginal Expected Shortfall) of bank $i$ at level $\alpha$ Acharya, Pedersen, Philippon & Richardson (2017)
  - $\mathrm{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$: the average loss of the bank on the worst $\alpha$ share of days for the market $m$
- Why "marginal": **if** $m = \sum_i w_i X_i$ is the portfolio itself, then $\mathrm{ES}_\alpha(X_m) = \sum_i w_i \, \mathrm{MES}_{i,\alpha}$
  - this is the Euler allocation (proof in the Appendix)
  - Our market indices are **not** portfolios of these 13 banks
  - Here MES measures **exposure**, not a share of the index ES
- Historical estimate: average $-X_i$ over the days with $X_m \leq \hat{q}_\alpha(X_m)$
  - With $\alpha = 5\%$ and 3 960 days, about 200 days

# MES 5% of Thirteen Banks (1/2)




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# MES 5% of Thirteen Banks (2/2)

- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026
  - 95% intervals: moving-block bootstrap, blocks of 20 days
- US banks: 3.22% (JPMorgan) to 4.04% (Morgan Stanley)
  - Europe up to 4.81% (ING)
  - Same days, market: 2.72% (S&P 500), 3.16% (Euro Stoxx 50): banks lose more
- HSBC is the exception in Europe (2.54%, below the index)
  - A London listing with business concentrated in Asia
- Banca Transilvania 3.14%, BRD 2.69%
  - BET itself 2.70%

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# From MES to a Crisis: LRMES

- **LRMES** (Long-Run MES): the expected fall of the bank's market equity in a crisis
  - $\mathrm{LRMES}_{i,t} = -\mathbb{E}_t\left[R_{i,t+\mathbf{1}:t+h} \mid R_{m,t+\mathbf{1}:t+h} < C\right]$, with $R$ the $h$-day arithmetic returns of bank and market
  - Crisis: the market falls below $C = -40\%$ over six months Acharya, Engle & Richardson (2012), or below $C = -10\%$ over $h = 22$ days Brownlees & Engle (2017)
- Shortcut of Acharya, Engle & Richardson (2012), without simulation: $\mathrm{LRMES}_i \approx 1 - \exp(-18 \times \mathrm{MES}_i^{\mathbf{2\%}})$
  - $\mathrm{MES}_i^{\mathbf{2\%}}$: average daily loss of the bank on days when the market falls by more than 2%
  - The 18 scales it to the six-month crisis, with a constant beta
  - Example: $\mathrm{MES}^{\mathbf{2\%}} = 3\%$ gives $1 - e^{-\mathbf{0.54}} = 41.7\%$
- On our banks: 37% (HSBC) to 58% (ING)
  - JPMorgan 48%, Banca Transilvania 52%

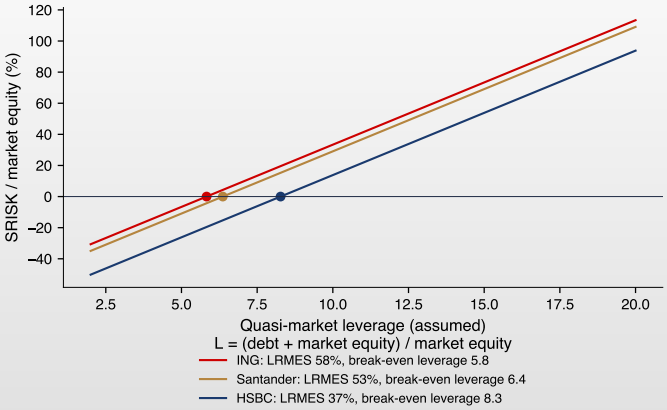
# SRISK and the Break-Even Leverage (1/2)

- **SRISK** (systemic risk): the capital a bank would lack in a severe market crisis
- Equity must be at least a share $k$ of quasi-assets, book debt $D$ plus market equity $W$ Brownlees & Engle (2017)
  - $\mathrm{SRISK}_i = k\,D_i - (1-k)(1-\mathrm{LRMES}_i)\,W_i$, assuming debt is not renegotiated in the crisis
  - Equivalently, $k[D_i + (1-\mathrm{LRMES}_i)W_i] - (1-\mathrm{LRMES}_i)W_i$
  - That is, the capital required after the crisis minus the capital left
- $k = 8\%$ for US firms Brownlees & Engle (2017) and $k = 5.5\%$ for European firms Engle, Jondeau & Rockinger (2015)
  - IFRS (International Financial Reporting Standards) balance sheets allow less **netting** of derivatives than US GAAP (Generally Accepted Accounting Principles)
  - So they report larger assets
- Positive SRISK is a capital shortfall
  - the system SRISK, the sum of the positive values, is published by V-Lab, New York University Stern School of Business

## SRISK and the Break-Even Leverage (2/2)

- Divide by $W$, with quasi-market leverage $\Lambda = (D + W)/W$: $\dfrac{\mathrm{SRISK}}{W} = k(\Lambda - 1) - (1 - k)(1 - \mathrm{LRMES})$
- SRISK is positive once $\Lambda$ exceeds the **break-even leverage** $\Lambda^* = 1 + \dfrac{(1 - k)(1 - \mathrm{LRMES})}{k}$
- Example: $D = 900$, $W = 100$ (so $\Lambda = 10$), $\mathrm{LRMES} = 41.7\%$
  - $\mathrm{SRISK} = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
  - $\Lambda^* = 7.7 < \Lambda = 10$: a shortfall
- Our data contain prices, not balance sheets
  - we report $\Lambda^*$ from the estimated LRMES
  - we treat leverage as an **assumption**

# SRISK as a Function of Leverage (1/2)




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# SRISK as a Function of Leverage (2/2)

- Banks with the highest, median and lowest LRMES
  - dots mark $\Lambda^*$
  - Leverage on the horizontal axis is assumed, not observed
- Curves in the chart, with $k = 8\%$ and the shortcut LRMES: $\Lambda^* = 5.8$ (ING), 6.4 (Santander), 8.3 (HSBC)
  - Other banks: JPMorgan 7.0, Morgan Stanley 6.0, Banca Transilvania 6.5, BRD 7.1
- Above $\Lambda^*$ the bank needs new capital in a 40% market crash
  - i.e. with debt above 4.8–7.3 times market equity
- With $k = 5.5\%$ the thresholds rise (ING 8.2, HSBC 11.9)
  - SRISK depends on the chosen prudential ratio
- Limits of the calculation
  - LRMES extrapolates daily data
  - market equity itself falls in a crisis

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# LRMES via GJR–GARCH and DCC: the Model

- Model of Brownlees & Engle (2017): zero-mean daily log returns of the bank and of the market
  - estimated by two-step quasi-maximum likelihood (QML)
- Volatility: GJR–GARCH(1,1) Glosten, Jagannathan & Runkle (1993)
  - GARCH: generalized autoregressive conditional heteroskedasticity
  - $\sigma_t^2 = \omega + (\alpha + \gamma\, 1\{r_{t-1} < 0\})\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$
  - $\gamma > 0$: extra reaction after a fall (leverage effect)
  - Here $\alpha$ is not the tail level
- Correlation: DCC(1,1) (dynamic conditional correlation) Engle (2002)
  - $Q_t = (1 - a - b)\, S + a\, \epsilon_{t-1} \epsilon_{t-1}^\top + b\, Q_{t-1}$, $\rho_t = Q_{12,t} / \sqrt{Q_{11,t} Q_{22,t}}$
  - $\epsilon_t = (r_{it}/\sigma_{it}, r_{mt}/\sigma_{mt})^\top$: standardised returns
  - $S$: their sample correlation matrix ($\bar{Q}$ in Engle (2002))
  - $a$: reaction and $b$: persistence of the correlation
  - $\rho_t$: today's bank–market correlation

# LRMES via GJR–GARCH and DCC: Simulation

- Simulation (details: Appendix)
  - Resample **pairs** $(\xi_{it}, \epsilon_{mt})$: the standardised market shock and the part of the bank shock uncorrelated with the market
  - Pairs keep the tail dependence that the Normal distribution misses
  - Filter the GARCH and DCC recursions $h$ days ahead from today's $\sigma^{\mathbf{2}}$ and $\rho$
  - LRMES: minus the average bank return over the crisis paths, those with $R_{m,t+\mathbf{1}:t+h} < C$
- Per market: 200 000 paths of 22 days ($C = -10\%$) and 400 000 of 126 days ($C = -40\%$)
  - data to 2026-09-18
- Probability of the 22-day crisis: today 1.3% (S&P 500), 4.3% (Euro Stoxx 50), 4.0% (BET)
  - against 12%, 19%, 8% on 31 March 2020

# LRMES by Simulation: Thirteen Banks (1/2)

| Bank | LRMES today | 90% range | LRMES 31.03.2020 | $\rho_T$ | $\Lambda^*$ (22 days) |
|---|---|---|---|---|---|
| JPMorgan | 10.6 | $[-2, 25]$ | 23.7 | 0.53 | 11.3 |
| Bank of America | 14.3 | $[-4, 35]$ | 23.2 | 0.44 | 10.9 |
| Citigroup | 13.0 | $[-1, 28]$ | 32.8 | 0.54 | 11.0 |
| Goldman Sachs | 16.5 | $[1, 34]$ | 25.0 | 0.65 | 10.6 |
| Morgan Stanley | 16.3 | $[1, 32]$ | 27.2 | 0.65 | 10.6 |
| Wells Fargo | 6.7 | $[-8, 24]$ | 22.9 | 0.32 | 11.7 |
| Deutsche Bank | 14.4 | $[2, 28]$ | 25.7 | 0.66 | 10.8 |
| BNP Paribas | 15.6 | $[2, 33]$ | 28.5 | 0.68 | 10.7 |
| Santander | 16.6 | $[3, 34]$ | 26.4 | 0.76 | 10.6 |
| ING | 12.5 | $[1, 27]$ | 32.9 | 0.68 | 11.1 |
| HSBC | 8.3 | $[-2, 19]$ | 10.9 | 0.58 | 11.5 |
| Banca Transilvania | 12.4 | $[-2, 34]$ | 17.2 | 0.74 | 11.1 |
| BRD-Groupe SG | 11.4 | $[-2, 28]$ | 15.8 | 0.64 | 11.2 |

- LRMES (%): market fall below $-10\%$ in 22 days Brownlees & Engle (2017)
  - 90% range: 5% and 95% quantiles of the bank's loss on the crisis paths; $\rho_T$: last DCC correlation
  - $\Lambda^*$ with $k = 8\%$ from this 22-day LRMES (not comparable with the shortcut $\Lambda^*$)

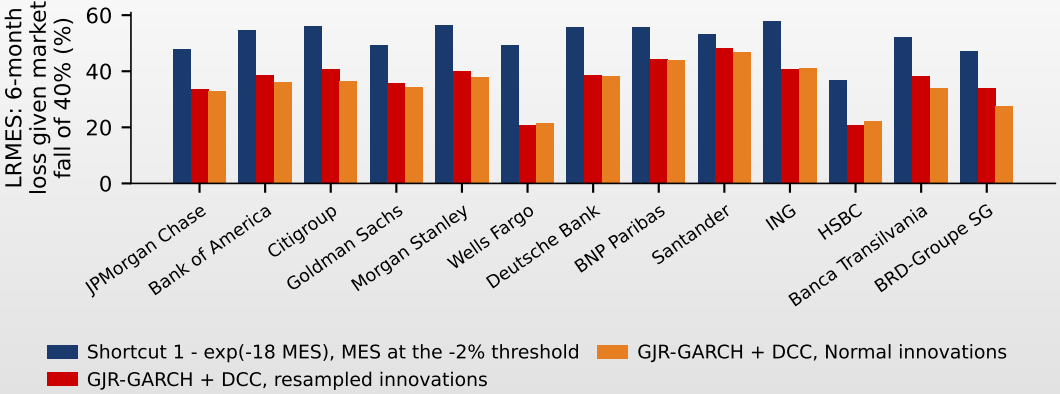
## LRMES by Simulation: Thirteen Banks (2/2)

- LRMES is **conditional**: it moves with today's volatility and correlation
  - JPMorgan: 10.6% today, 23.7% with the information of 31 March 2020
  - Deutsche Bank: 14.4% and 25.7%
- The 90% range is wide Brownlees & Engle (2017)
  - LRMES is an average over crisis paths
  - The shortfall in one particular crisis can be much larger
  - The range describes scenario dispersion, not uncertainty about the estimated mean
- The scenario, not only the bank, sets the level
  - One-month, $-10\%$ crisis: LRMES about a third of the six-month, $-40\%$ numbers of Acharya, Engle & Richardson (2012)

## Reading the LRMES Table: Ranges and Scenarios

- A negative lower end of the 90% range is not a sign error
  - A negative path loss is a gain: on at least 5% of the crisis paths the bank's shares rise while the market falls
  - It happens when the bank is weakly correlated with its market today, e.g. Wells Fargo: $\rho_T = 0.32$, range $[-8, 24]$
- $\Lambda^*$ inherits the crisis scenario through LRMES, so compare it only within one scenario
  - example: JPMorgan, $k = 8\%$
  - Shortcut, six months, market $-40\%$: LRMES 48%, $\Lambda^* = 7.0$
  - Simulation, 126 days, market $-40\%$: LRMES 33%, $\Lambda^* = 8.7$
  - Simulation, 22 days, market $-10\%$: LRMES 10.6%, $\Lambda^* = 11.3$

# Model Risk: LRMES under Three Specifications (1/2)




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# Model Risk: LRMES under Three Specifications (2/2)

- Six-month crisis, market $-40\%$ Acharya, Engle & Richardson (2012)
  - shortcut $1 - e^{-18\,\mathrm{MES}^{2\%}}$ vs GJR–GARCH + DCC simulation with resampled or Normal innovations
- Simulated LRMES over the shortcut: 0.42 to 0.90
  - rank correlation 0.77
  - JPMorgan: 48% (shortcut), 33% (resampled), 33% (Normal)
  - ING: 58%, 41%, 41%
  - The shortcut uses the whole history
  - The simulation starts from today's volatility
- Resampled vs Normal innovations: rank correlation 0.95
  - Levels differ most for Citigroup (40.8% vs 36.3%) and BRD (34.1% vs 27.5%)
- Same data, different defensible models, different numbers: model risk Danielsson, James, Valenzuela & Zer (2016)

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## How Much Worse Is the System When a Bank Is in Distress?

- MES looks from the market to the bank
  - **contribution** looks the other way, from the bank to the system
- We need a quantile in the tail of the system, not its mean
  - Quantile regression estimates the conditional quantile directly Koenker & Bassett (1978)
- Levels: $\alpha = 1\%$ and $\alpha = 5\%$, those reported by Adrian & Brunnermeier (2016)
  - we compare the rankings at the two levels
- Questions of this section
  - How much does each bank add to the tail loss of the system?
  - Are the differences between banks statistically significant?

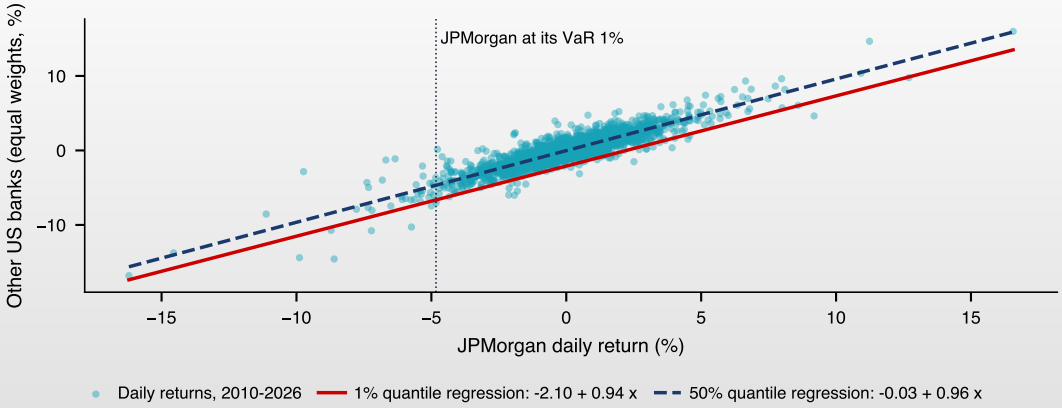
## CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank $i$ at its VaR Adrian & Brunnermeier (2016)
  - $P\left(X^{sys} \leq -\mathrm{CoVaR}_{\alpha}^{sys|i} \;\middle|\; X_i = -\mathrm{VaR}_{i,\alpha}\right) = \alpha$
  - The loss of the system that is exceeded with probability $\alpha$ on a day when bank $i$ is at its own VaR
- **$\Delta$CoVaR**: the change from the median state of the bank, a normal day, to its distress state
  - $\Delta\mathrm{CoVaR}_{\alpha}^{sys|i} = \mathrm{CoVaR}_{\alpha}^{sys|X_i=q_\alpha} - \mathrm{CoVaR}_{\alpha}^{sys|X_i=q_{50\%}}$, with the bank return fixed at the quantile shown
- $\Delta$CoVaR measures the **contribution** of bank $i$ to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**
  - the VaR of bank $i$ when the system is in distress

# Estimating CoVaR by Quantile Regression

- Quantile regression Koenker & Bassett (1978): $q_\alpha(Y \mid x) = a_\alpha + b_\alpha x$
  - $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg\min_{a,b} \sum_t \rho_\alpha(y_t - a - bx_t)$
  - The **check function** $\rho_\alpha(u) = u\,(\alpha - 1\{u < 0\})$ weights negative residuals by $1 - \alpha$ and positive ones by $\alpha$
- 1. Regress the system on the bank: $\hat{q}_\alpha(X^{sys} \mid X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- 2. Plug in the bank's distress quantile: $\mathrm{CoVaR}_\alpha^{sys|i} = -\left(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i)\right)$
- 3. $\Delta\mathrm{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha\,(\hat{q}_{\mathbf{50\%}}(X_i) - \hat{q}_\alpha(X_i))$
  - the intercept cancels
- Uncertainty: moving-block bootstrap with 999 resamples
  - it keeps volatility clustering
  - The block length grows like $n^{\mathbf{1/3}}$ Hall, Horowitz & Jing (1995): $3\,960^{\mathbf{1/3}} \approx 16$, rounded to 20 days, one trading month

# Quantile Regression: JPMorgan and the Other US Banks (1/2)



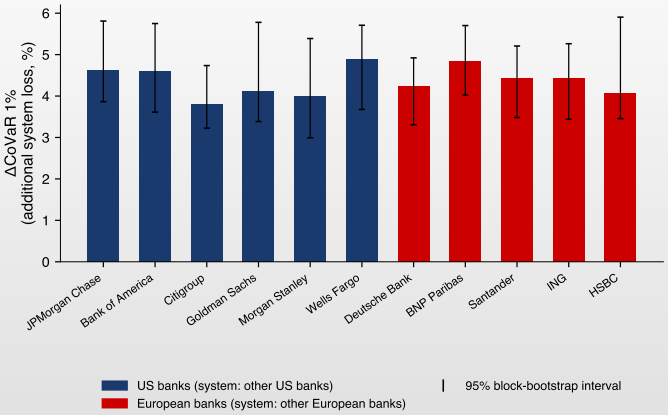

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# Quantile Regression: JPMorgan and the Other US Banks (2/2)

- Daily returns 2010–2026, 1% and 50% quantile regression lines
  - dotted line: JPMorgan at its VaR 1%
- 1% line: $\hat{q}_{1\%} = -2.10 + 0.94\, X_i$
  - median line: $-0.03 + 0.96\, X_i$
  - Similar slopes: the 1% line is shifted down, not steeper
- With JPMorgan at its VaR 1% (4.83%), the other US banks have CoVaR 1% = 6.64%
  - Against the system VaR 1% of 5.07% and the CoVaR 1% of 2.03% with JPMorgan at its median
- $\Delta\mathrm{CoVaR}_{1\%} = 4.62\%$
  - JPMorgan in distress adds about 4.62 points to the tail loss of the system
  - Association, not causation: a common shock hits all banks on the same day

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# ΔCoVaR 1% of Eleven Banks (1/2)




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# ΔCoVaR 1% of Eleven Banks (2/2)

- Bars: ΔCoVaR, the **extra** system loss when the bank moves from its median to its VaR
  - with 95% bootstrap intervals
  - No Romanian banks: with only two, the "system" of the other bank would be one single bank
- From 3.81% (Citigroup) to 4.89% (Wells Fargo)
  - intervals 1.5–2.4 points wide
  - The marginal intervals overlap, e.g. Citigroup $[3.23, 4.74]$ and Wells Fargo $[3.68, 5.71]$
- Overlap is **not** a test
  - estimates from the same days are strongly correlated
  - so the interval for the difference can be narrower
- At $\alpha = 5\%$ the values are smaller (JPMorgan 2.62%) and the ranking changes
  - Spearman correlation of the 1% and 5% rankings: 0.10, a fragile basis for regulation

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# Testing ΔCoVaR 1%

- All tests use the same 999 block-bootstrap draws for every bank
  - so the estimates keep their dependence (details: Appendix)
- (1) No contribution, $H_0: b_\alpha = 0$
  - Wald test with the sandwich variance of quantile regression Koenker & Bassett (1978), estimated with a kernel Hendricks & Koenker (1992)
  - $t$-ratios of $\hat{b}_{1\%}$ from 4.0 (HSBC) to 6.9 (JPMorgan): every bank contributes (Quantile Regression: Asymptotics)
- (2) $H_0: b_\alpha = b_{50\%}$, rejected for 2 of 11 banks (Citigroup, ING, also after Holm)
  - not rejecting for the other 9 does not prove equality
- (3) Equal contributions, $H_0: \Delta\mathrm{CoVaR}_i = \Delta\mathrm{CoVaR}_j$
  - dominance tests: Castro & Ferrari (2014); results by bank: Appendix
  - At $\alpha = 1\%$: 4 of 25 within-region pairs differ at 5% (0 after Holm); joint Wald $p = 0.015$ (US), 0.31 (Europe)
  - At $\alpha = 5\%$: 3 of 25 pairs differ (0 after Holm); joint Wald $p = 0.014$ (US), 0.046 (Europe)

# Reading the Tests: Holm and the Joint Wald Test

- **Many pairs, one family**: 25 pairs tested at 5%
  - more than one false "difference" is expected even with equal contributions
  - Holm (1979): sort $p_{(1)} \leq \cdots \leq p_{(m)}$ and reject while $p_{(k)} \leq 0.05/(m-k+1)$
  - The family-wise error stays below 5% for any dependence
  - Toy check, $p = 0.010, 0.030, 0.040$: $0.010 \leq 0.05/3$ (reject), $0.030 > 0.05/2$ (stop): one rejection instead of three
- **Joint Wald test** of equal contributions within a region
  - $W = \hat{d}^\top \hat{V}^{-1} \hat{d} \sim \chi^2$ with the number of banks minus 1 degrees of freedom
  - $\hat{d}$: differences from one reference bank (any choice gives the same $W$)
  - $\hat{V}$: their bootstrap covariance
- US: the joint test rejects ($p = 0.015$) while 0 of 25 pairs survive Holm
  - The joint test pools moderate differences; Holm needs one very strong pair ($p_{(1)} \leq 0.05/25$)
  - So the contributions are not all equal, but we do not know which pair differs

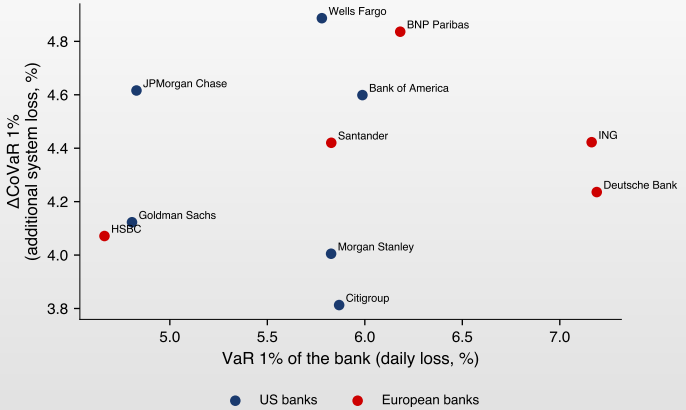
# Losses at Least Equal to VaR: the Girardi–Ergün CoVaR

- Girardi & Ergün (2013) condition on the bank's loss being **at least** its VaR
  - $P\left(X^{sys} \leq -\mathrm{CoVaR}_{\alpha}^{\leq} \,\middle|\, X_i \leq -\mathrm{VaR}_{i,\alpha}\right) = \alpha \iff P\left(X^{sys} \leq -\mathrm{CoVaR}_{\alpha}^{\leq}, X_i \leq -\mathrm{VaR}_{i,\alpha}\right) = \alpha^2$
  - Benchmark state of $\Delta\mathrm{CoVaR}^{\leq}$: $|X_i - \mu_i| \leq \sigma_i$ instead of the median
  - Below we compare levels only
- It rises with the strength of dependence, which the CoVaR given $X_i = -\mathrm{VaR}_{i,\alpha}$ need not do Mainik & Schaanning (2014)
- It can be **backtested**: the joint hit has probability $\alpha^2$
- Historical estimate at $\alpha = 5\%$, on the 198 days with $X_i \leq \hat{q}_{5\%}(X_i)$
  - JPMorgan 7.70% vs 3.81% by quantile regression
  - eleven banks: 7.3–8.2% vs 3.6–4.8%
  - Under the Normal distribution this change raises CoVaR only modestly
  - The factor of two points to departures from the Normal benchmark, without identifying their cause

# A Normal Check: Which CoVaR Rises with Dependence?

- Bivariate Normal with zero means, $\sigma_{sys} = 1.5$, $\alpha = 1\%$, $z = -2.326$
  - the bank sits at its VaR, $X_i = -\mathrm{VaR}_{i,\alpha}$
  - Given the bank, the system has mean $\rho\,\sigma_{sys} z$ and standard deviation $\sigma_{sys}\sqrt{1-\rho^2}$
  - Hence $\mathrm{CoVaR}_\alpha = \sigma_{sys}|z|\left(\rho + \sqrt{1-\rho^2}\right)$ and, with the median state $X_i = 0$, $\Delta\mathrm{CoVaR}_\alpha = \rho\,\sigma_{sys}|z|$
- From $\rho = 0.6$ to $\rho = 0.9$: $\mathrm{CoVaR}_{1\%}$ falls from 4.89% to 4.66%
  - because $\rho + \sqrt{1-\rho^2}$ peaks at $\rho = 0.71$
  - $\Delta\mathrm{CoVaR}_{1\%}$ rises from 2.09% to 3.14%
- Lesson: a CoVaR level conditioned on $X_i = -\mathrm{VaR}_{i,\alpha}$ can fall when banks become more interdependent
  - while $\Delta$CoVaR and $\mathrm{CoVaR}^{\leq}$ rise with $\rho$ Mainik & Schaanning (2014)

# VaR and ΔCoVaR Measure Different Things (1/2)




MFM_ch18_covar

# VaR and ΔCoVaR Measure Different Things (2/2)

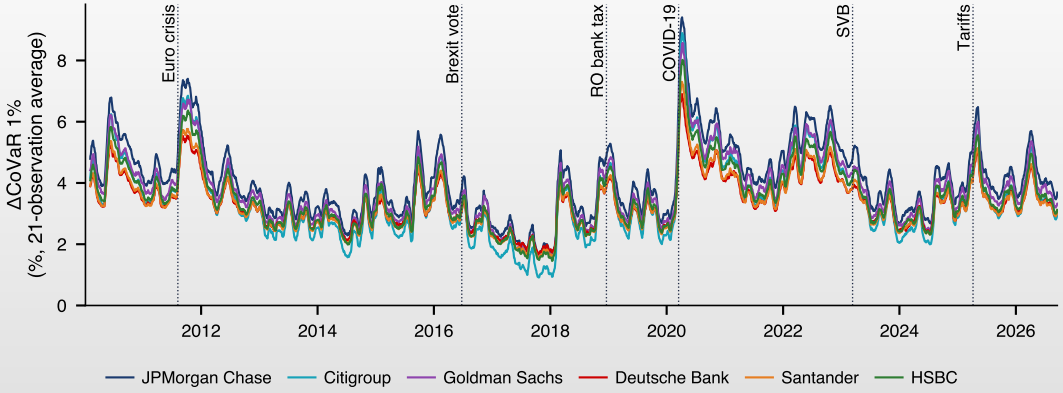
- Horizontal axis: VaR 1% of the bank, vertical axis: its ΔCoVaR 1%
  - eleven banks, 2010–2026
- Spearman rank correlation between VaR and ΔCoVaR: 0.16
  - Deutsche Bank and ING: the largest VaR (7.19%, 7.16%), only average ΔCoVaR
  - JPMorgan: one of the smallest VaR (4.83%), one of the largest ΔCoVaR
- Central message of Adrian & Brunnermeier (2016)
  - Regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals the correlation is very uncertain

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# ΔCoVaR 1% over Time: State Variables

- The question: does the contribution of a bank rise in stress periods?
- The quantiles of the bank depend on state variables $M_{t-1}$ Adrian & Brunnermeier (2016)
  - $M_{t-1}$: log VIX, regional index return and system return, all lagged one day
  - $\Delta\mathrm{CoVaR}_{\alpha,t} = \hat{b}\left(\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i)\right)$, with $\hat{q}_{\cdot,t}$ conditional on $M_{t-1}$ (Appendix)
- The slope $\hat{b}$ stays constant: ΔCoVaR moves with the conditional VaR of the bank

# ΔCoVaR 1% over Time (1/2)



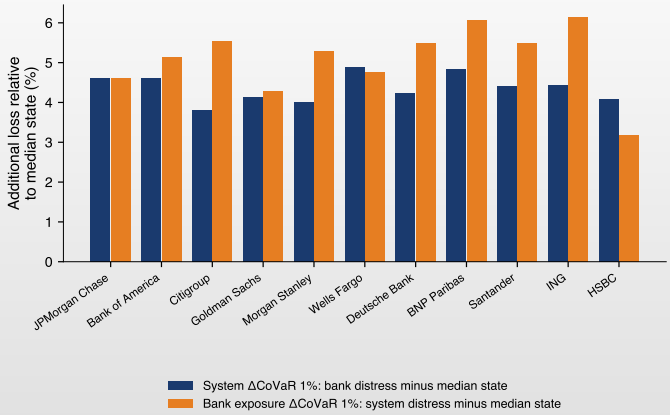

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# ΔCoVaR 1% over Time (2/2)

- Six banks, moving averages over 21 observations of the daily ΔCoVaR 1%
- JPMorgan: mean 2.16% in 2017, mean 9.06% in March 2020
  - peak 10.99% on 2020-03-18
  - Deutsche Bank: mean 2.04% in 2017, mean 6.65% in March 2020
- Low in calm periods, exactly when risk may build up in balance sheets
  - Hence Adrian & Brunnermeier (2016) propose a forward ΔCoVaR, predicted from leverage, size and maturity mismatch
  - Maturity mismatch: long-term assets funded by short-term debt, which must be rolled over

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# Contribution versus Exposure (1/2)




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## Contribution versus Exposure (2/2)

- Blue, contribution: loss of the system when the bank is in distress
- Orange, exposure: loss of the bank when the system is in distress
  - $X_i$ regressed on $X^{sys}$, quantiles of $X^{sys}$ plugged in
- The direction matters: ING has contribution 4.42% and exposure 6.14%
  - BNP Paribas 4.84% and 6.07%
- HSBC: contribution 4.07%, exposure only 3.17%
- Exposure exceeds contribution for most European banks
- Both are associations, not causal effects
- Exposure matters for the bank's own capital
  - contribution matters for the regulator

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## How Much Can We Trust a Ranking?

- Rankings by different systemic measures disagree and carry wide uncertainty Nucera, Schwaab, Koopman & Lucas (2016)
  - combine measures and report rank uncertainty
- Model risk Danielsson, James, Valenzuela & Zer (2016)
  - simulated vs shortcut LRMES
  - Girardi–Ergün vs quantile-regression CoVaR
  - $\alpha = 1\%$ vs 5%
- Effective tail sample $\alpha n$: 40 days at 1%, 198 at 5%
  - At 1% the kernel and block-bootstrap standard errors disagree (Appendix)
- Out-of-sample backtests of $\mathrm{CoVaR}^{\leq}_{5\%}$ reject for no bank (Appendix)
  - With about 7.4 expected joint hits their power is low
- What survives are large, tested differences
  - across regions and in the tail slope of some banks
  - not the order of eleven close point estimates

## Through Which Links Do Shocks Travel between Banks?

- MES and CoVaR look at one bank and one aggregate
  - a crisis, however, travels between many banks at once
- A network answers three questions
  - Who transmits shocks and who receives them?
  - How connected is the system as a whole?
  - Does connectedness rise in crises?
- The approach: the forecast-error variance decomposition of a VAR Diebold & Yilmaz (2014)
  - on the daily volatilities of the banks

# From Pairs to Networks: the Variance Decomposition (1/2)

- A **network** looks at all pairs of banks at once
  - Nodes: banks; directed edges: how much of the future uncertainty of bank $i$ comes from shocks to bank $j$
- Vector autoregression (VAR) of order $p$ for $N$ series $y_t$
  - $y_t = c + \sum_{l=1}^{p} A_l\, y_{t-l} + \varepsilon_t$, $\mathrm{Cov}(\varepsilon_t) = \Sigma$
  - Each series is regressed on $p$ lags of all series
  - $c$: constants; $A_l$: $N \times N$ coefficient matrices
  - $\varepsilon_t$: shocks; $\Sigma$: their covariance matrix (assumptions: Appendix)
- Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- Other networks
  - Granger causality Billio, Getmansky, Lo & Pelizzon (2012) (Appendix)
  - tail networks TENET (Tail-Event driven NETwork risk) Härdle, Wang & Yu (2016)
  - the Financial Risk Meter (next section)

# From Pairs to Networks: the Variance Decomposition (2/2)

- The $H$-day forecast error of $y_i$ is a sum of shocks to all series
  - so its variance can be split by source
  - Moving-average form $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$: $\Phi_h$ is the response of $y$ after $h$ days to a shock today, $\Phi_0 = I$
- Orthogonal (Cholesky) shocks depend on the ordering of the series
  - the **generalized** decomposition does not Koop, Pesaran & Potter (1996), Pesaran & Shin (1998)
  - $\theta_{ij}(H) = \dfrac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$, normalised: $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$
  - Numerator: the part of the forecast error variance of $i$ due to a shock in $j$
  - Denominator: the total forecast error variance of $i$
  - $e_i$: the $i$-th unit vector (it picks series $i$)
  - $\sigma_{jj}$: the variance of shock $j$, the $j$-th diagonal element of $\Sigma$

## The Connectedness Table

- $\tilde{\theta}_{ij}$ (in %): the share of the $H$-day forecast error variance of $i$ due to shocks in $j$ Diebold & Yilmaz (2014)
  - Each row sums to 100; before normalisation it does not, because generalized shocks are correlated
- **From** others: $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$
- **To** others: $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
- **Net**: $T_i - F_i$, positive for a net transmitter
- **Pairwise net**: $\tilde{\theta}_{ji} - \tilde{\theta}_{ij} > 0$ means $i$ transmits net to $j$
  - drawn as an arrow $i \to j$
- **Total connectedness**: $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$
  - the average share of variance that comes from the others

# Example: the Connectedness Table for Two Banks

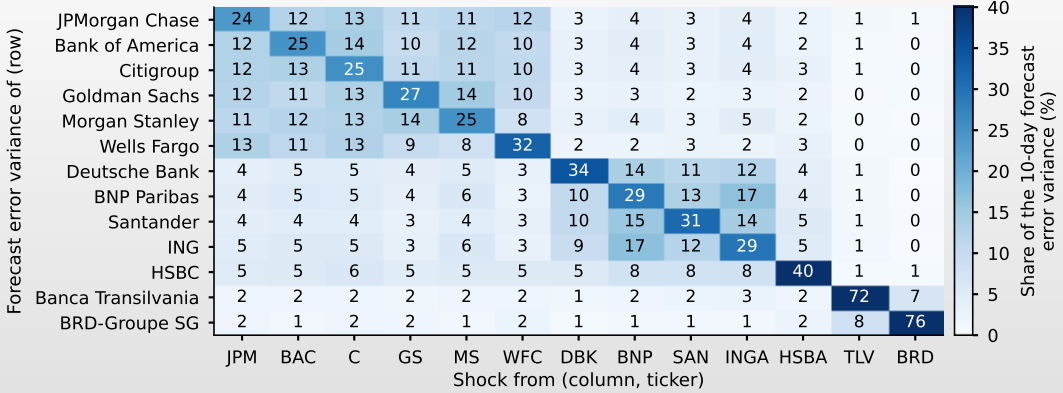
| Bank $i \setminus$ shock $j$ | 1 | 2 | $F_i$ |
|---|---|---|---|
| 1 | 80 | 20 | 20 |
| 2 | 40 | 60 | 40 |
| $T_j$ | 40 | 20 | |

- Toy normalised table $\tilde{\theta}_{ij}$ (%)
  - row $i$: the bank whose variance is split
  - column $j$: the source of the shock
  - rows sum to 100
- Off-diagonal cells only
  - From others: $F_1 = 20$, $F_2 = 40$ (row sums)
  - To others: $T_1 = 40$, $T_2 = 20$ (column sums)
- Net: bank 1 $40 - 20 = +20$ (transmitter), bank 2 $-20$
  - net positions always sum to zero
- Total: $C = (20 + 40)/2 = 30\%$

# Parkinson Range Volatility and the VAR of the Thirteen Banks

- **Parkinson volatility** from the daily high price $P_t^{\max}$ and low price $P_t^{\min}$ Parkinson (1980)
  - $\hat{\sigma}_t^{\mathbf{2}} = \dfrac{(\ln P_t^{\max} - \ln P_t^{\min})^{\mathbf{2}}}{4 \ln 2}$, annualised $100\sqrt{252\,\hat{\sigma}_t^{\mathbf{2}}}$, used in logs
  - Unbiased for the variance of a Brownian motion without drift and more efficient than the squared return
- Our model: VAR on the log volatility of the 13 banks, a range-based volatility as in Demirer et al. (2018)
  - Volatility, not returns: it jumps in crises, so its network shows where stress travels
  - Order by the Bayesian information criterion (BIC): $p = 2$
  - Horizon $H = 10$ days, as in Diebold & Yilmaz (2012)
  - US banks close after European ones; the VAR lags capture next-day transmission

# The Connectedness Table of Thirteen Banks (1/2)



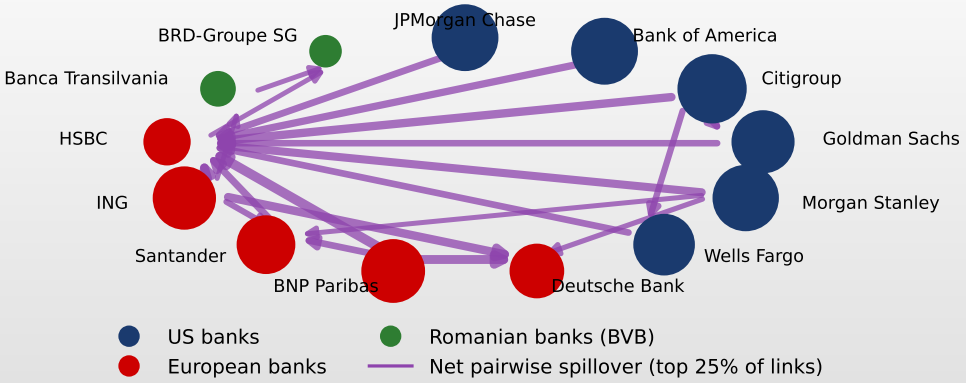

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# The Connectedness Table of Thirteen Banks (2/2)

- Row $i$, column $j$: share (%) of the 10-day forecast error variance of bank $i$ due to shocks in bank $j$
  - full sample (tickers of the data table)
- Total connectedness $C = 63.9\%$
  - on average two thirds of a bank's volatility risk comes from the other banks
  - On returns instead of volatility: $C = 75.4\%$
- Two regional blocks
  - a US bank receives 84.0% from US shocks
  - a European bank 72.7% from European shocks
  - Across the Atlantic: 26.2% from US to a European bank, 15.0% the other way
- Romanian banks: mostly their own shocks
  - 72.3% (Banca Transilvania) and 76.1% (BRD)
  - From US banks 10.3%, from European banks 8.0%

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# The Net Spillover Network (1/2)




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# The Net Spillover Network (2/2)

- Arrow $j \to i$ when $j$ transmits net to $i$ (largest 25% of pairwise net spillovers)
- Node size grows with the spillover to others
- Net transmitters: Citigroup (+19.0), Bank of America (+10.9), Morgan Stanley (+9.9), JPMorgan (+9.8)
- Net receivers: HSBC (−23.6), BRD (−13.0), Deutsche Bank (−12.6), Banca Transilvania (−12.2)
  - HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years

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# Connectedness over Time (1/2)




MFM_ch18_spillover_rolling

# Connectedness over Time (2/2)

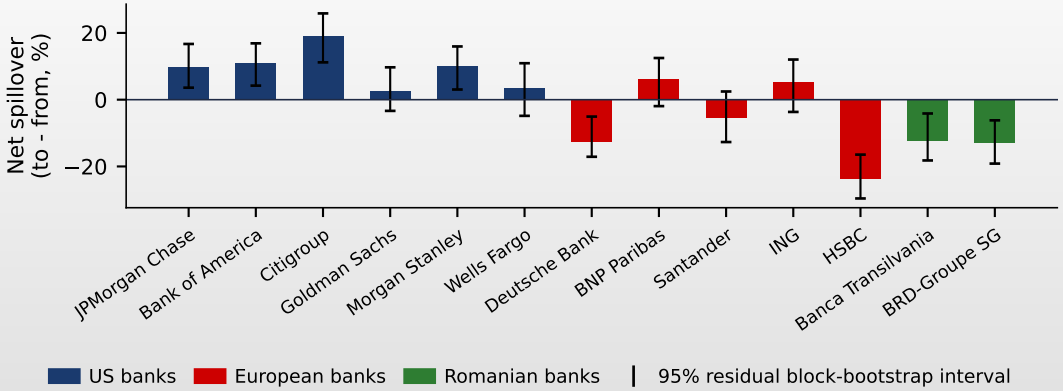
- Top: total connectedness on rolling 250-day windows
  - VAR(2), $H = 10$, every 5 days
- Bottom: net spillover summed by region
- The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27)
  - mean 62.0%
  - COVID-19: from 61.8% in 2019 to 78.4% in spring 2020
  - After the 2023 bank failures 66.0%, in the calm year 2024 53.5%
  - The index rises within days
  - Its fall about 250 days later partly reflects the crisis leaving the window
- US banks are net transmitters in 89% of the windows
  - sum of their net spillovers, on average $+48.4$ points
- European banks were net transmitters during the euro crisis of 2011–2012
  - and mostly net receivers afterwards
- Romanian banks are net receivers on average ($-24.1$ points on average)

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## Estimation Error: How Reliable Is the Network?

- The question: which "transmitter" and "receiver" labels survive estimation error?
- A VAR(1) on 13 series has 169 slope coefficients
  - the VAR(2) used here has $2 \times 169 = 338$
  - A 250-day window: 250 observations per equation for 26 slopes and a constant
  - The table is a nonlinear function of all of them
- Residual block bootstrap, 999 draws
  - Resample 20-day blocks of residual vectors, rebuild the series with the estimated VAR, re-estimate, recompute the table

# Estimation Error: Bootstrap Intervals for the Network (1/2)




MFM_ch18_spillover_uncertainty

# Estimation Error: Bootstrap Intervals for the Network (2/2)

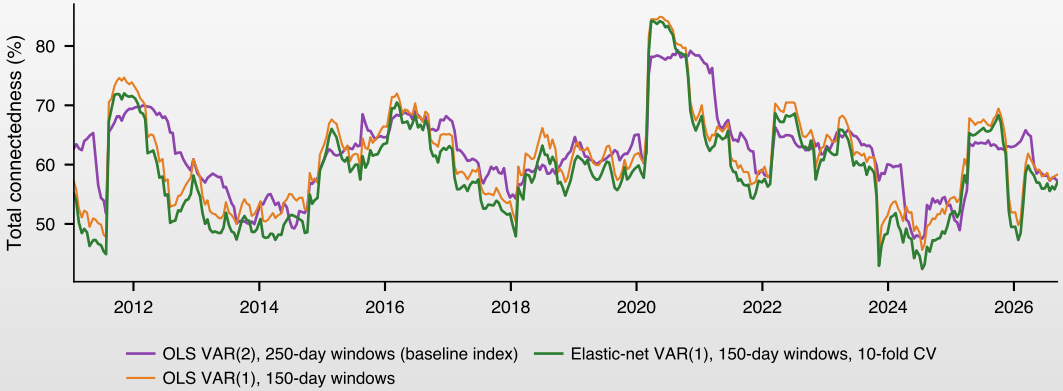
- Full sample, $p = 2$: $C = 63.9\%$, 95% interval $[59.9, 65.8]$
  - Peak window (to 2020-08-27): $C = 79.3\%$, $[65.9, 83.0]$
  - Bootstrap mean 75.8%: small samples also bias $C$
- 8 of 13 net positions have intervals that exclude zero
  - For the other 5 the label "transmitter" or "receiver" is not identified
  - The net spillover network draws the largest 25% of links, some of them not significant
  - Hence the intervals

MFM_ch18_spillover_uncertainty

# Elastic-Net Connectedness: Does the Network Depend on the Estimator?

- A VAR(2) on 13 series has 338 slopes: in short windows OLS overfits
- Demirer et al. (2018): each VAR equation by the adaptive elastic net
  - in the 150-day windows, the plain elastic net
  - $\lambda$ chosen equation by equation by 10-fold cross-validation, "10-fold CV" in the chart (definition in the Appendix)
- We compare, on the same window end dates, the base index with indices on 150-day windows

# OLS and Elastic-Net Connectedness in Rolling Windows (1/2)




MFM_ch18_spillover_uncertainty

# OLS and Elastic-Net Connectedness in Rolling Windows (2/2)

- Full sample: $C = 63.3\%$ (OLS, ordinary least squares: 63.9%)
  - 59% of coefficients non-zero
- Means over 372 common window end dates, moving by 10 trading days
  - baseline recomputed on those same dates
  - Baseline (OLS VAR(2), 250 days) 62.0%
  - OLS VAR(1), 150 days 61.1%; elastic-net VAR(1), 150 days 58.7%
- The dynamics agree: correlation 0.82 between the elastic-net and the base index
  - crisis peaks in the same weeks
  - Levels depend on window, estimator and lag order
  - Changes over time are robust

MFM_ch18_spillover_uncertainty

## How Tight Are the Links between Banks in the Tail?

- The spillover network describes an average shock, while systemic risk sits in the tail
- Financial Risk Meter (FRM) Mihoci, Althof, Chen & Härdle (2020): one quantile regression at $\tau = 5\%$ for each firm $j$
  - its return on **all** other firms and on lagged macro variables
  - With 100 firms and 63 days there are more regressors than days
  - The LASSO penalty keeps only the strong links
  - The FRM is the average selected penalty, an indicator of tail interdependence and systemic stress
- The question of this section: does this indicator track systemic stress on our 13 banks too?

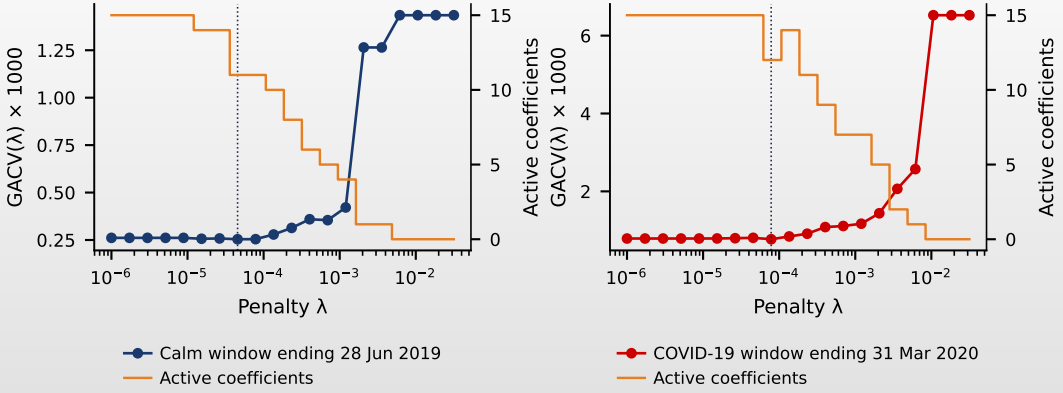
# The Financial Risk Meter and the GACV Penalty (1/2)

- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression Belloni & Chernozhukov (2011)
  - $\hat{\beta}_j(\lambda) = \arg\min_\beta \frac{1}{n} \sum_t \rho_\tau \left( X_{j,t} - \beta_{\mathbf{0}} - \beta^\top Z_{-j,t} \right) + \lambda \|\beta\|_{\mathbf{1}}$, over the $n$ days of the window
  - $X_{j,t}$: return of firm $j$
  - $Z_{-j,t}$: returns of the other firms and the lagged macro variables
  - A larger $\lambda$ leaves fewer active coefficients
- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients
  - $\mathrm{GACV}(\lambda) = \frac{\sum_t \rho_\tau \left( X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \mathrm{df}(\lambda)}$, $\hat{X}_{j,t}(\lambda)$: fitted quantile; $\mathrm{df}$: number of nonzero coefficients
  - The selected $\lambda_j$ minimises GACV
  - The FRM is the average of the $\lambda_j$

# The Financial Risk Meter and the GACV Penalty (2/2)

- The FRM was built for US financial firms Mihoci, Althof, Chen & Härdle (2020)
  - and for emerging markets Ben Amor, Althof & Härdle (2022) and China Wang, Althof & Härdle (2023)
  - Extensions: FRM based on expectiles Ren, Lu, Li & Härdle (2022), for cryptocurrencies Ren, Althof & Härdle (2025) and for the EU energy sector Toma, Dima, Grecu, Pele & Lessmann (2026)
- Our version: 13 banks
  - macro variables S&P 500, Euro Stoxx 50 and VIX at $t-1$
  - 63-day windows every 10 days
  - $\tau = 5\%$ and 63-day windows, about one trading quarter, are those of Mihoci, Althof, Chen & Härdle (2020)
  - 15 regressors for 63 days: far fewer than in the published FRM

# GACV in a Calm and in a Crisis Window (1/2)



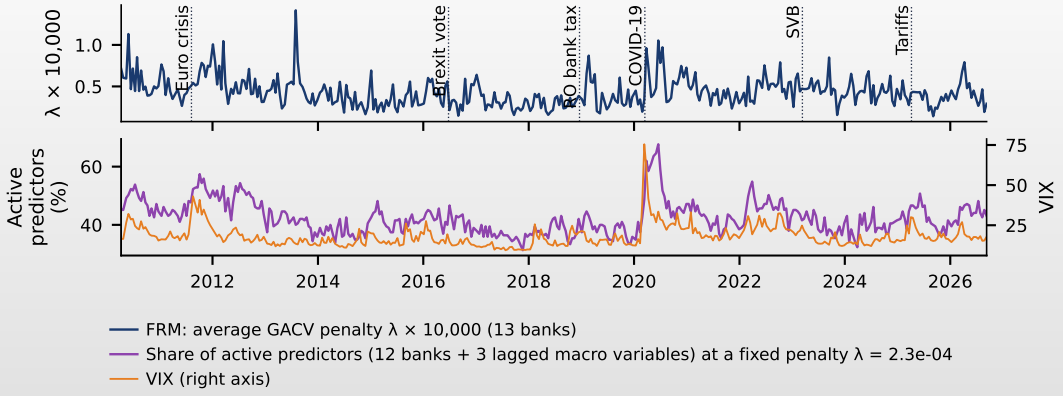

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# GACV in a Calm and in a Crisis Window (2/2)

- JPMorgan on the other 12 banks and three lagged macro variables, 63 days
  - Dotted line: selected $\lambda$; orange: active coefficients
- Calm window (to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active
  - COVID-19 window (to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active
- In the crisis the whole GACV curve is about three times higher
  - The check loss is in return units, and crisis returns are larger
- With few regressors the minimum lies at almost no penalty: the selection is noisy

MFM_ch18_frm

# The Financial Risk Meter of Thirteen Banks (1/2)




MFM_ch18_frm

# The Financial Risk Meter of Thirteen Banks (2/2)

- Top: average GACV penalty over 13 banks
  - 390 windows of 63 days
  - Bottom: share of active predictors at a fixed $\lambda$, with the VIX
- The average GACV penalty is noisy: correlation with the VIX only 0.35
- At a fixed $\lambda = 2.3 \times 10^{-4}$ the share of active predictors follows the VIX more closely (0.62)
  - From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: on this small panel, neither measure is shown to be a reliable systemic-risk indicator
  - The fixed-penalty count is only an exploratory sensitivity check

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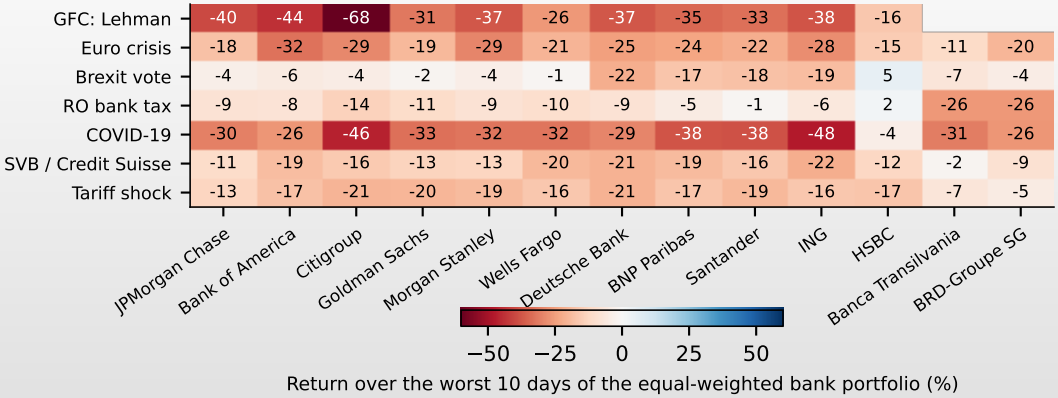
# How Much Would We Lose in a Severe but Plausible Scenario? (1/2)

- A **stress test** applies a severe but plausible scenario to the banks and measures the loss
  - **Historical**: replay a past crisis
  - **Hypothetical**: shock chosen factors
  - **Reverse**: find the most plausible scenario behind a given loss (e.g. one that would make the bank non-viable)
  - Supervisory tests: Federal Reserve stress tests (US), European Banking Authority (EU)
  - Global systemically important banks: a regulatory list, scored by the Basel Committee
  - Score: size, interconnectedness, substitutability, complexity and cross-border activity
  - These banks hold extra capital Basel Committee (2018)

# How Much Would We Lose in a Severe but Plausible Scenario? (2/2)

- Seven episodes, each a window of a few weeks around a known shock
  - Lehman (autumn 2008; "GFC" in the chart: global financial crisis), euro crisis (summer 2011), Brexit vote (June 2016), Romanian bank tax (December 2018)
  - COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- In each window: the **10 consecutive trading days with the lowest cumulative return**
  - cumulative return of the equal-weighted bank portfolio
  - We report every bank's return over that interval
  - Ten days match the market-risk horizon (Chapter 7)
  - 2008 uses the 11 international banks only: the Romanian series start in 2010

# Seven Historical Stress Scenarios (1/2)




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# Seven Historical Stress Scenarios (2/2)

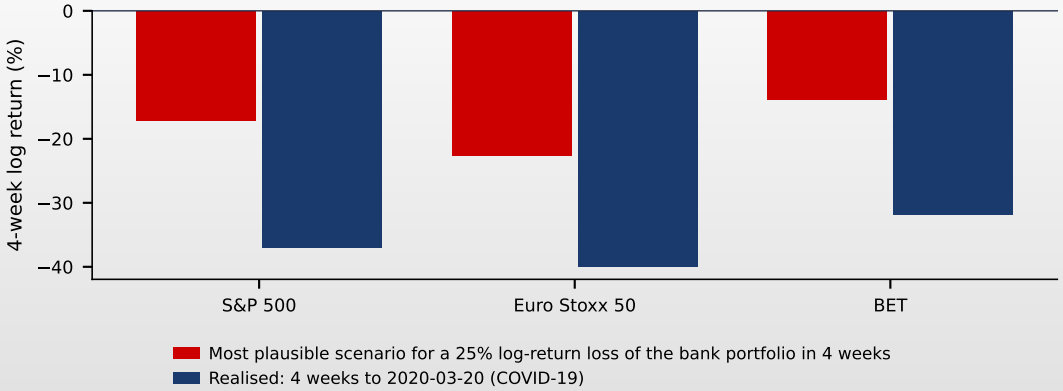
- Simple return (%) of each bank over the worst 10 consecutive days of the bank portfolio in each episode
  - $100(e^{\sum_t r_t/100} - 1)$; blank: no data
- Portfolio loss: Lehman 37.6% (2008-11-10 – 2008-11-21)
  - COVID-19 31.9%, euro crisis 22.3%
  - Tariffs 16.0%, SVB 14.8%, Romanian bank tax 10.2%, Brexit 7.9%
- Worst single cells: Citigroup −68% in 2008 (−114% as a cumulative log return)
  - ING −48% in March 2020
- Shocks are local as well as global: Brexit hit mainly European banks
- The Romanian bank tax hit Romanian banks hardest (−26%, −26%)
  - US banks lost between 8% and 14%
- A stress test built only on global crises would miss the domestic political shock

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## Reverse Stress Testing

- Factor model of weekly bank log returns (%): $X_i = \beta_i^\top f + e_i$
  - factors $f$ = (S&P 500, Euro Stoxx 50, BET)
  - Over $h$ weeks $f \sim N(0, h\Sigma)$, $\Sigma$: weekly factor covariance, no serial covariance
- Loss of the equal-weighted portfolio of the 13 banks: $L = -b^\top f$ (log loss in %)
  - $b$: the average of the $\beta_i$
  - residual $\sum_i w_i e_i$ set to zero
- Most plausible scenario with loss $\ell^*$
  - minimise the squared Mahalanobis distance $f^\top (h\Sigma)^{-\mathbf{1}} f$ subject to $-b^\top f = \ell^*$
  - $f^* = -\ell^* \dfrac{h\Sigma\, b}{b^\top h\Sigma\, b}, \qquad d^* = \dfrac{\ell^*}{\sqrt{b^\top h\Sigma\, b}}$
  - $d^*$: the target loss in portfolio standard deviations
  - Larger means less plausible (two-factor example: ▸ Appendix)

# The Most Plausible Scenario for a 25% Log-Return Loss (1/2)




MFM_ch18_stress

# The Most Plausible Scenario for a 25% Log-Return Loss (2/2)

- Weekly data 2010–2026 (871 weeks), four-week horizon, the length of the acute phase of March 2020
  - Red: $f^*$ for $\ell^* = 25\%$
  - Blue: factor moves in the worst four weeks of 2020
- $b = (0.45, 0.65, 0.18)$
  - four-week standard deviation of the portfolio 5.79%
  - A 25% log loss (arithmetic 22.1%) needs S&P 500 −17.2%, Euro Stoxx 50 −22.6%, BET −13.9% together
- $d^* = 4.31$: Normal probability $8.0 \times 10^{-6}$ per four weeks
  - Yet in the four weeks to 2020-03-20 the portfolio had a log loss of 52.0% (factor model: 48.5%)
- The shape of the scenario is informative, its probability is not
  - heavy tails and rising correlations make Normal probabilities far too small

MFM_ch18_stress

# Climate Risk: Physical and Transition Risk



Seat of the European Central Bank, Frankfurt, 2015

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

- **Physical risk**: losses from floods, droughts, heat and storms
  - these events damage collateral and borrowers
- **Transition risk**: losses from the move to a low-carbon economy
  - carbon prices, regulation, technology
- "Green swan" risks: potentially extremely disruptive events
  - they could be behind the next systemic financial crisis
  - backward-looking risk models cannot anticipate them Bolton et al. (2020)

# The ECB Climate Stress Test of 2022: Results

- 104 significant banks took part in the ECB test of 2022
  - a bottom-up projection of losses covered 41 of them ECB Banking Supervision (2022)
  - bottom-up: each bank runs the supervisor's scenarios on its own data and models
- About 60% of banks had no climate risk stress-testing framework
- Only 20% considered climate risk when granting loans
- Credit and market losses of around EUR 70 billion for the 41 banks
  - scenarios: a short-term disorderly transition and two physical risk scenarios
  - The ECB states that this significantly understates the risk
  - Reasons: data gaps, rudimentary models, no second-round effects

## Limits of Stress Tests

- Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- Second-round effects are hard to model and usually left out
  - fire sales, funding runs, counterparty losses
  - as the ECB noted for its climate test
- Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move
  - balance-sheet stress tests look forward but slowly
- Early-warning research combines market measures with balance-sheet data
  - e.g. entropy-based measures of market risk Pele, Lazar & Dufour (2017)

## Does the Network Change with the Size of Shocks?

- Ando, Greenwood-Nimmo & Shin (2022): does the topology of a financial network change with the **size** of shocks?
  - Their version of the question of Diebold & Yilmaz (2014): how much of the future uncertainty of $i$ is due to idiosyncratic shocks from $j$, as the shock size varies?
- Why it matters: the connectedness of this chapter is estimated at the conditional mean
  - OLS, LASSO, elastic net Demirer et al. (2018)
  - i.e. it describes how an average shock spreads
  - Systemic shocks are larger than average
  - Reading a rise in rolling connectedness as a crisis signal assumes that mean relations also hold in the tails
- Link to this chapter: the quantile regression of CoVaR Adrian & Brunnermeier (2016), Koenker & Bassett (1978)
  - now inside the VAR and its variance decomposition
  - Giglio, Kelly & Pruitt (2016) showed that systemic risk shifts the lower tail of future output more than its centre
  - Ando, Greenwood-Nimmo & Shin (2022) ask the same tail-versus-centre question of the network itself
- Application: the credit risk loop between sovereigns and their banks
  - bailouts move bank risk onto the state

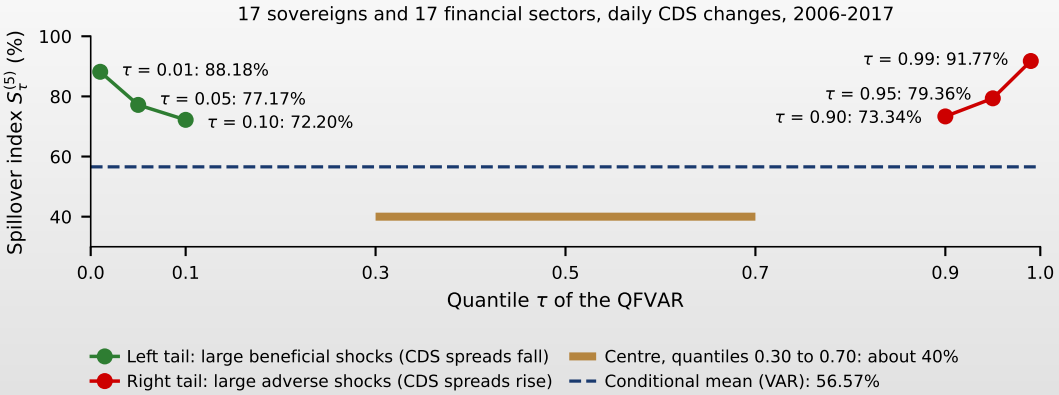
# CDS Data and the Factor Quantile VAR

- CDS (credit default swap) spread: the annual premium paid for protection against default
  - default of a sovereign or bank; premium in basis points of the insured amount (1 bp = 0.01%)
  - it rises when default risk rises, so an increase is bad news
- Data: daily first differences of 5-year CDS spreads (basis points)
  - 17 sovereigns and 17 national financial sectors, $m = 34$
  - 3 January 2006 – 29 December 2017, 3 129 days
- Quantile factor VAR (QFVAR): $y_t = \mu + \sum_{j=1}^{p} \Phi_j y_{t-j} + \Gamma f_t + v_t$
  - each equation fitted at quantile $\tau$ by minimising the check loss
  - $y_t$: the $m$ CDS changes; $\mu$: constants; $\Phi_j$: VAR coefficient matrices; $\Gamma$: factor loadings
  - Unobserved factors $f_t$ absorb common shocks, so $v_t$ is idiosyncratic
  - Their number by the criterion of Ando & Bai (2020): 10 at the mean

## Spillovers and Adverse Credit Shocks

- Generalized forecast error variance decomposition (FEVD) of the idiosyncratic shocks at quantile $\tau$
  - rows normalised to 1
  - spillover index $S_\tau^{(h)} = m^{-1} \sum_i F_{i\leftarrow\bullet,\tau}^{(h)}$
  - $F_{i\leftarrow\bullet,\tau}^{(h)}$: the share of the $h$-day forecast error variance of $i$ that comes from the others
  - It is the "from" of the connectedness table; $S_\tau^{(h)}$ is its average over the $m$ series
- Rolling windows of $w = 250$ days (2 880 windows)
  - number of factors fixed at its full-sample value (full specification: Appendix)
- Relative tail dependence $\mathrm{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$
  - For CDS changes the right tail ($\tau = 0.95$) holds large adverse shocks, the left tail ($\tau = 0.05$) large beneficial ones

# Spillovers across Quantiles (1/2)



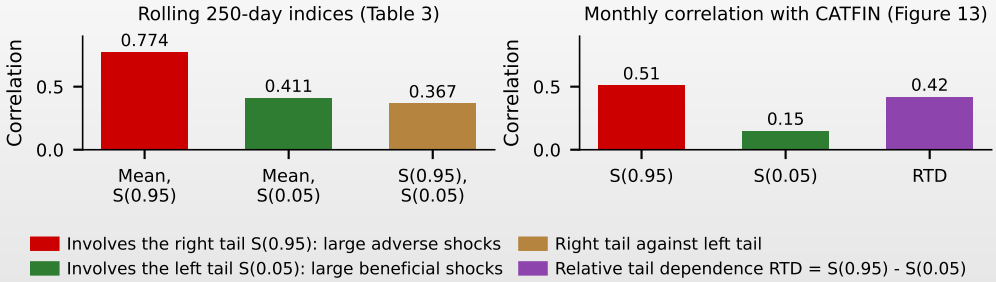

MFM_ch18_quantile_connectedness

# Spillovers across Quantiles (2/2)

- Full-sample spillover index $S_\tau^{(5)}$ at each quantile $\tau$
  - vs the conditional-mean index (source: Ando, Greenwood-Nimmo & Shin (2022))
- U-shaped profile: about 40% in the centre, 88.18% at $\tau = 0.01$, 91.77% at $\tau = 0.99$
  - The mean index (56.57%) describes neither regime: too high for small shocks, too low for large ones
- Full sample: almost symmetric tails, $\mathrm{RTD} = 79.36 - 77.17 = 2.19$ points
  - in rolling windows the asymmetry is stronger (next chart)
- For stress testing: mean connectedness understates spillovers in a crisis
  - it understates how much of a sector's forecast error variance comes from other sectors

MFM_ch18_quantile_connectedness

# The Tails Move Apart (1/2)




- Left: correlations between the rolling indices ($w = 250$ days, $h = 5$)
- Right: monthly correlations with CATFIN (source: Ando, Greenwood-Nimmo & Shin (2022))
- CATFIN: the average of three VaR 1% estimates for financial stocks Allen, Bali & Tang (2012)

MFM_ch18_quantile_connectedness

## The Tails Move Apart (2/2)

- The mean index follows the adverse tail far more than the beneficial tail
  - correlation 0.774 with $S_{0.95}$, against 0.411 with $S_{0.05}$
  - A rise in mean connectedness is mainly a rise in bad-news spillovers
- The two tails share little variation (0.367)
  - Two indicators, not one index measured twice
  - Hence RTD is informative: about $\pm 15$ points in rolling windows, against 2.19 on the full sample
- Against CATFIN: 0.51 for $S_{0.95}$, 0.42 for RTD, only 0.15 for $S_{0.05}$
  - CATFIN moves with adverse-shock spillovers more than with beneficial ones (weak bilateral links and the timing of RTD: Appendix)

# Extension to Our 13 Banks: Specification and Open Questions

- Course data: daily closes of the 13 banks of this chapter, 2010–2026
  - course dataset; US, euro area and United Kingdom, Romania
  - $y_t$ = daily loss $-r_t$ in %, so that the right tail is the adverse one, as for CDS
  - Data limitations: equity losses instead of CDS spreads
  - 13 banks and no sovereigns ($m = 13$ instead of 34)
  - Different closing times of US, European and Romanian markets
- Limits stated by the authors
  - the lag order of the mean is imposed at every quantile
  - the quantile is held fixed along the horizon (other limits: Appendix)
- Open extension
  - Does the RTD of the 13 banks rise before March 2020 and the March 2023 bank failures?
  - Does it predict bank stress beyond current volatility and the mean-based index of this chapter?
  - For forecasting, select lag order and factors on a training sample only

# Does Connectedness Warn of Crises, or Only Track Volatility?

## An open research question: systemic risk or only volatility?

- Connectedness indices are monitored as early-warning tools
  - but an index from a VAR on volatilities may co-move with volatility without adding predictive information
  - Live research: regularised VAR networks Demirer et al. (2018); tail connectedness differs from mean connectedness Ando, Greenwood-Nimmo & Shin (2022)
- Formal version: $y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}$, $H_0 : \beta_C = 0$, $H_1 : \beta_C > 0$
  - $y_{t+h}$: future bank stress; $C_t$: total connectedness; $\sigma_t$: current volatility
  - Evidence against $H_1$: the confidence interval (CI) for $\beta_C$ rules out economically relevant effects, not mere non-significance
- Identification: an edge is a share of forecast error variance, not a causal exposure
  - A common shock raises volatility and connectedness together, with no bank infecting another

# The Discovery Loop, Step by Step (1/2)

- **Literature**: map what is known, then verify every reference
  - Resolve each DOI (Digital Object Identifier) on Crossref and check that the paper really controls for volatility
  - Search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses**: competing readings of a rise in connectedness, each with its own test
  - E.g. a common shock fades with volatility
  - Contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code**: on the course data, replicating a known number first
  - Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter

# The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the predictive regressions are run
  - Every cell of the grid is reported
  - The family is corrected with Holm or controlled with the FDR (false discovery rate)
- **Critique**: ask the assistant to attack the result
  - Expect: add current volatility as a control, then test the increment of $R^2$ and of the out-of-sample loss
- **Report**: standardised effects with HAC $t$-statistics, every cell of the grid, what failed
  - HAC: heteroskedasticity- and autocorrelation-consistent
  - The prompts that shaped the design go into `AI_USE.md` (reusable prompts and the researcher's checks: Appendix)

# Mini-Case: Is Connectedness More Than Volatility? (1/2)

- Typical assistant proposal: "high total connectedness warns of future bank stress"
  - Tested on 731 weekly dates, 2011–2026
  - $y_{t+h}$: log mean Parkinson volatility of the 13 banks over the next $h = 63$ days
  - Control: the same over the last 20 days
  - Standardised variables; Newey–West with 13 lags, because the 63-day target overlaps across about 13 weeks
- Connectedness alone: $\hat{\beta}_C = 0.40$ ($t = 4.63$), $R^2 = 0.16$
  - the proposal looks right

# Mini-Case: Is Connectedness More Than Volatility? (2/2)

- With current volatility: $\hat{\beta}_C = 0.08$ ($t = 0.75$), $\hat{\gamma} = 0.53$ ($t = 4.13$)
  - $R^2 = 0.343$ vs 0.339 with volatility alone
  - Correlation of $C_t$ with volatility 0.59
  - Next 63-day maximum drawdown: $\hat{\beta}_C = -0.10$ ($t = -1.32$) even as the only regressor
- Interpretation: given current volatility, connectedness adds no significant predictive content
  - Not proof of zero: in our sample the persistence of volatility explains most of the predictive content
  - Network information may still matter in the tails Ando, Greenwood-Nimmo & Shin (2022) or for the direction of transmission

# Your Turn: a Project Seed (1/2)

- **Question**
  - Does **tail** connectedness predict bank distress beyond volatility, out of sample?
- **Replicate first**
  - Our rolling index: mean 62.0%, maximum 79.3% (2020-08-27)
  - The six-group connectedness table of Demirer et al. (2018) for 2003–2014: directional entries and row/column totals
  - Explain the deviations; the authors publish the replication data (journaldata.zbw.eu)
- **Extend**
  - Quantile connectedness at $\tau = 5\%$ Ando, Greenwood-Nimmo & Shin (2022)
  - Out-of-sample evaluation from 2018
  - MES and SRISK as alternative targets

# Your Turn: a Project Seed (2/2)

- **Deliverables**
  - A grid of specifications fixed in writing before the regressions
  - A notebook that regenerates every number
  - HAC inference and out-of-sample losses for every cell
- **AI log** (templates)
  - `AI_USE.md`: the prompts that shaped the design
  - `AI_ERRORS.md`: each wrong claim (a VAR on price levels, OLS $t$-statistics on overlapping windows) and the test that caught it

# Which Bank Is Most Systemic? (1/2)

| Bank | VaR 1% | MES 5% | ΔCoVaR 1% | Net spillover | $\Lambda^*$ |
|---|---|---|---|---|---|
| JPMorgan | 4.83 | 3.22 | 4.62 | 9.8 | 7.0 |
| Bank of America | 5.99 | 3.85 | 4.60 | 10.9 | 6.2 |
| Citigroup | 5.87 | 4.03 | 3.81 | 19.0 | 6.0 |
| Goldman Sachs | 4.81 | 3.34 | 4.12 | 2.4 | 6.8 |
| Morgan Stanley | 5.83 | 4.04 | 4.00 | 9.9 | 6.0 |
| Wells Fargo | 5.78 | 3.32 | 4.89 | 3.6 | 6.8 |
| Deutsche Bank | 7.19 | 4.54 | 4.24 | −12.6 | 6.1 |
| BNP Paribas | 6.18 | 4.53 | 4.84 | 6.0 | 6.1 |
| Santander | 5.83 | 4.23 | 4.42 | −5.4 | 6.4 |
| ING | 7.16 | 4.81 | 4.42 | 5.2 | 5.8 |
| HSBC | 4.66 | 2.54 | 4.07 | −23.6 | 8.3 |
| Banca Transilvania | 5.06 | 3.14 | – | −12.2 | 6.5 |
| BRD-Groupe SG | 4.42 | 2.69 | – | −13.0 | 7.1 |

- Daily losses in %
  - net spillover of volatility: to minus from
  - $\Lambda^*$: break-even leverage with $k = 8\%$ and the shortcut LRMES (six-month crisis, market $-40\%$)
  - –: no ΔCoVaR for the two Romanian banks, whose "system" would be a single bank

## Which Bank Is Most Systemic? (2/2)

- Rank correlations across the eleven international banks
  - MES and $\Delta$CoVaR $-0.01$
  - MES and net spillover 0.04
  - $\Delta$CoVaR and net spillover $-0.01$
  - By contrast, VaR and MES: 0.88
- No single ranking
  - supervisors combine several indicators with size and balance-sheet data, as in the Basel Committee (2018) methodology
- Romanian banks have a moderate MES
  - they are net receivers of volatility
  - their break-even leverage is close to that of US banks

## Key Takeaways

- Systemic risk lives in the joint tail
  - exposure (MES, SRISK), contribution ($\Delta$CoVaR), connectedness (spillover and tail networks)
- Euro-area banks have the largest MES in our sample (4.2–4.8%)
- US banks are the main net transmitters of volatility
- VaR and $\Delta$CoVaR rank banks differently, and rankings are uncertain
  - test differences, report intervals, state $\alpha n$
- Market-based measures jump in crises
  - stress tests and balance sheets are needed to see risk building up
- Romanian banks: mostly their own shocks
  - they receive more volatility spillover than they transmit

## Common Pitfalls

- Reading ΔCoVaR or a spillover edge as causal
  - a common shock creates the same pattern
- Reading overlapping intervals as "equal" and separate ones as "different"
  - Test the difference on joint draws and correct for the number of pairs (Holm and the joint Wald test)
- Including the bank itself in the system portfolio
  - its own return adds a mechanical self-contribution
- Orthogonal variance decompositions whose results depend on the order of the banks
- Reporting SRISK without balance sheets, or the $1 - e^{-18\mathrm{MES}}$ shortcut as the published measure
- Drawing net labels without intervals

# MES as an Euler Contribution: Proof (1/2)

- Portfolio $X_p = \sum_i w_i X_i$ with a continuous distribution
  - $\mathrm{ES}_\alpha(X_p) = -\mathbb{E}[X_p \mid X_p \leq q_\alpha(X_p)]$
  - $\mathrm{ES}_\alpha$ is positively homogeneous of degree 1 in $w$, so by Euler's theorem $\mathrm{ES}_\alpha(X_p) = \sum_i w_i \, \partial \mathrm{ES}_\alpha / \partial w_i$
  - Write $\mathrm{ES}_\alpha = -\frac{1}{\alpha}\mathbb{E}\left[X_p 1\{X_p \leq q_\alpha(w)\}\right]$, with $f_p$ the density of $X_p$
  - The derivative has an interior and a boundary term:
    $$\frac{\partial}{\partial w_i}\mathbb{E}[X_p 1\{X_p \leq q_\alpha\}] = \mathbb{E}[X_i 1\{X_p \leq q_\alpha\}] + q_\alpha f_p(q_\alpha)\left(\frac{\partial q_\alpha}{\partial w_i} - \mathbb{E}[X_i \mid X_p = q_\alpha]\right)$$
  - The quantile derivative is $\partial q_\alpha / \partial w_i = \mathbb{E}[X_i \mid X_p = q_\alpha]$ Gouriéroux, Laurent & Scaillet (2000), so the boundary term is zero
  - Hence $\partial \mathrm{ES}_\alpha / \partial w_i = -\mathbb{E}[X_i \mid X_p \leq q_\alpha(X_p)] = \mathrm{MES}_{i,\alpha}$ with $m = p$ Gouriéroux, Laurent & Scaillet (2000), Scaillet (2004)

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# MES as an Euler Contribution: Proof (2/2)

- Check on our data: equal-weighted portfolio of the six US banks, $\alpha = 5\%$ (198 tail days)
  - $\mathrm{ES}_{5\%}$ of the portfolio $= 4.32\% =$ average of the six MES conditional on the portfolio (4.32%)
  - The identity is exact, also for the historical estimator
  - Conditioning on the S&P 500 instead gives an average MES of 3.63% (JPMorgan: 3.22% vs 3.85%)
  - The identity fails when $m \neq p$
- Under the bivariate Normal distribution with zero means, $\mathrm{MES}_{i,\alpha} = \beta_i \, \mathrm{ES}_\alpha(X_m)$: a beta times a constant

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# LRMES Simulation: Innovations and Recursions

- Standardised residuals $\epsilon_{it} = r_{it}/\sigma_{it}$, $\epsilon_{mt} = r_{mt}/\sigma_{mt}$
  - $S$ is their sample correlation matrix, the long-run target of $Q_t$
  - Bank innovation orthogonal to the market: $\xi_{it} = (\epsilon_{it} - \rho_t\,\epsilon_{mt})/\sqrt{1-\rho_t^2}$ Brownlees & Engle (2017)
- Each simulated day $s$: draw a past date $\tau$ and take the pair $(\xi_{i\tau}, \epsilon_{m\tau})$
  - rebuild the bank shock with the **simulated** correlation
  - $\epsilon_{i,t+s} = \rho_{t+s}\,\epsilon_{m\tau} + \sqrt{1-\rho_{t+s}^2}\,\xi_{i\tau}$, then update $\sigma^2$ and $Q$ with the new returns
- Restrictions, so that $\sigma_t^2 > 0$ and $Q_t$ is positive definite
  - GJR–GARCH: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$, $\alpha + \gamma/2 + \beta < 1$ (finite variance for symmetric innovations)
  - in general, finite variance needs $\alpha + \gamma\,\mathbb{E}[\epsilon^2 1\{\epsilon < 0\}] + \beta < 1$
  - DCC: $a, b > 0$, $a + b < 1$
  - Initialisation: $Q_1 = S$ in estimation; the simulation starts from $\sigma^2_{T+1}$ and $Q_{T+1}$ filtered to the last date

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# Quantile Regression: Asymptotics

- Asymptotics for a fixed $\alpha$, with $\hat{\theta}_\alpha = (\hat{a}_\alpha, \hat{b}_\alpha)^\top$
  - $n$ days, $x_t = (1, X_{i,t})^\top$, $f_t$: the conditional density of $Y$ at its $\alpha$-quantile
  - $\sqrt{n}(\hat{\theta}_\alpha - \theta_\alpha) \Rightarrow N\left(0,\ \alpha(1-\alpha)\, D_{\mathbf{1}}^{-\mathbf{1}} D_{\mathbf{0}} D_{\mathbf{1}}^{-\mathbf{1}}\right)$, $D_{\mathbf{0}} = \mathbb{E}[x_t x_t^\top]$, $D_{\mathbf{1}} = \mathbb{E}[f_t\, x_t x_t^\top]$
  - Assumes independent observations (or scores $x_t(\alpha - 1\{u_t < 0\})$ that are a martingale difference), $f_t > 0$ near the quantile and $D_{\mathbf{0}}$, $D_{\mathbf{1}}$ positive definite
  - With serially dependent scores $D_{\mathbf{0}}$ needs a long-run (HAC) estimate or a block bootstrap
  - The **sparsity** $1/f_t$ must be estimated (kernel or bandwidth)
  - At $\alpha = 1\%$ few observations lie near the quantile, so this estimate is noisy
- The slope $b_\alpha$ can change with $\alpha$
  - the bank may matter more for the system in the tail than in the centre

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# Extremal Quantiles: Why $\alpha = 1\%$ Is Hard

- At $\alpha = 1\%$ only $\alpha n = 40$ days remain in the tail
  - The Normal approximation above assumes $\alpha n \to \infty$ fast; with $\alpha n$ of a few dozen it can be poor
  - Extremal-quantile theory (extreme-value limits, subsampling) is the alternative built for this case Chernozhukov & Fernández-Val (2011)
- In our data the block-bootstrap standard error of $\Delta$CoVaR 1% is about 0.6 times the kernel-based one
  - median over banks
  - the two methods give clearly different uncertainty
  - The kernel-based standard error rests on a sparsity estimate from the few observations near the 1% quantile
  - The bootstrap redraws the same few tail days: neither is reliable when $\alpha n$ is this small
- Practical rule: report $\alpha = 5\%$ and $\alpha = 1\%$ with $\alpha n$
  - expect the 1% ranking to be the noisier one
  - The low Spearman correlation between the 1% and 5% rankings (0.10) may reflect estimation noise, genuine differences between the two tail levels, or both

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# Testing $\Delta$CoVaR: Three Hypotheses

- (1) $H_0: b_\alpha = 0$ (no contribution)
  - Tests of linear hypotheses across quantiles use the joint asymptotics of $(\hat{b}_\alpha, \hat{b}_{50\%})$ Koenker & Bassett (1982)
- (2) $H_0: b_\alpha = b_{50\%}$ (tail link equal to the central link)
  - We estimate the variance of $\hat{b}_\alpha - \hat{b}_{50\%}$ from 999 moving-block bootstrap draws that keep the pairs of days together
- (3) $H_0: \Delta\mathrm{CoVaR}_i = \Delta\mathrm{CoVaR}_j$ (equal contributions)
  - $z_{ij} = (\widehat{\Delta\mathrm{CoVaR}}_i - \widehat{\Delta\mathrm{CoVaR}}_j)/\widehat{\mathrm{se}}^*$ for each pair, with $\widehat{\mathrm{se}}^*$ the bootstrap standard error of the difference
  - Holm (1979) correction over the pairs; a joint Wald test that all are equal

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# Testing ΔCoVaR 1%: Results

| Bank | ΔCoVaR | $\hat{b}_{1\%}$ | $t$ | $\hat{b}_{50\%}$ | $\hat{b}_{1\%} - \hat{b}_{50\%}$ | $p$ |
|---|---|---|---|---|---|---|
| JPMorgan | 4.62 | 0.94 | 6.9 | 0.96 | −0.02 | 0.66 |
| Bank of America | 4.60 | 0.76 | 6.5 | 0.78 | −0.02 | 0.73 |
| Citigroup | 3.81 | 0.64 | 4.7 | 0.76 | −0.11 | $< 0.001$ |
| Goldman Sachs | 4.12 | 0.85 | 5.6 | 0.84 | 0.00 | 0.99 |
| Morgan Stanley | 4.00 | 0.68 | 4.5 | 0.73 | −0.05 | 0.28 |
| Wells Fargo | 4.89 | 0.84 | 4.3 | 0.78 | 0.06 | 0.17 |
| Deutsche Bank | 4.24 | 0.59 | 4.1 | 0.60 | −0.02 | 0.67 |
| BNP Paribas | 4.84 | 0.77 | 6.5 | 0.74 | 0.03 | 0.37 |
| Santander | 4.42 | 0.75 | 4.5 | 0.69 | 0.07 | 0.07 |
| ING | 4.42 | 0.61 | 4.4 | 0.71 | −0.10 | 0.002 |
| HSBC | 4.07 | 0.87 | 4.0 | 0.86 | 0.01 | 0.96 |

- $t$: $\hat{b}_{1\%}$ divided by its kernel (sandwich) standard error, valid only for serially independent scores
- $p$: $p$-value of $b_{1\%} = b_{50\%}$
  - from $z = (\hat{b}_{1\%} - \hat{b}_{50\%})/\widehat{\mathrm{se}}^*$, with the bootstrap standard error $\widehat{\mathrm{se}}^*$ and the Normal approximation
- pair tests and joint Wald tests: main slide

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# Time-Varying $\Delta$CoVaR with State Variables

- State variables $M_{t-1}$ Adrian & Brunnermeier (2016)
  - log VIX, the return of the regional index and of the system, all lagged one day
  - $q_\alpha(X_{i,t} \mid M_{t-1}) = a_i + c_i^\top M_{t-1}$ at $\alpha$ and at 50% ($a_i$, $c_i$: intercept and coefficients of the bank's quantile regression)
  - $q_\alpha(X_t^{sys} \mid X_{i,t}, M_{t-1}) = a + b\,X_{i,t} + c^\top M_{t-1}$
- Then $\Delta\mathrm{CoVaR}_{\alpha,t} = \hat{b}\left(\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i)\right)$
  - It moves with the bank's conditional VaR, which moves with the VIX
- The slope $\hat{b}$ is constant: time variation comes from the state, not from the link

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# Backtesting CoVaR and MES (1/2)

- Systemic measures are conditional risk measures, so they need **conditional** backtests Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)
  - For $\mathrm{CoVaR}^{\leq}$: hit $I_t = 1\{X_t^{sys} \leq -\mathrm{CoVaR}_t^{\leq},\ X_{i,t} \leq -\mathrm{VaR}_{i,t}\}$ with $P(I_t = 1) = \alpha^2$ under $H_0$
  - This holds if both the bank VaR and the conditional quantile are calibrated
  - The Kupiec test checks that the hit frequency is $\alpha^2$
  - With $x$ hits in $n$ days and $\hat{\pi} = x/n$, $\mathrm{LR}_{\mathrm{POF}} = -2\ln \dfrac{(1-\alpha^2)^{n-x}(\alpha^2)^x}{(1-\hat{\pi})^{n-x}\hat{\pi}^x} \sim \chi_1^2$ under $H_0$ (Chapter 8)
  - The Christoffersen test also checks that hits are independent: a hit today must not make one tomorrow more likely
  - The bank VaR is checked separately, against its own rate $\alpha$
  - For MES: the cumulative joint violation process of Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)

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# Backtesting CoVaR and MES (2/2)

- Our out-of-sample check: historical $\mathrm{CoVaR}^{\leq}_{5\%}$ on an expanding window
  - from day 1 000, 2 960 forecast days per bank
  - Expected joint hits 7.4; observed 5–9 across the eleven banks (JPMorgan 9)
  - Smallest Kupiec $p = 0.35$: no rejection
  - Power is limited: if the true joint rate were twice $\alpha^2$, the 5% Kupiec test would reject in 62% of samples of this length with independent hits
  - So even a doubling is missed about one time in three

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# VAR Assumptions and the Lag Order of the Rolling Index

- The VAR($p$) of From Pairs to Networks
  - Assumptions: stable VAR (roots of $\det(I - \sum_l A_l z^l)$ outside the unit circle), $\varepsilon_t$ serially uncorrelated with finite, positive definite $\Sigma$
  - Moving-average form: $y_t = \mu + \sum_{h \geq \mathbf{0}} \Phi_h \varepsilon_{t-h}$, $\Phi_{\mathbf{0}} = I$, $\Phi_h = \sum_{l=\mathbf{1}}^{p} A_l \Phi_{h-l}$ ($\Phi_h = 0$ for $h < 0$)
  - $e_i$: the $i$-th unit vector
- Lag order: every window keeps $p = 2$, the order BIC selects on the full sample
  - in the windows BIC picks $p = 1$ every time
  - with $p = 1$ the index has correlation 0.99 with the one shown

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# Regularised VARs: the Elastic Net

- Demirer et al. (2018): each VAR equation by the adaptive elastic net
  - $\lambda$ chosen equation by equation by 10-fold cross-validation
  - $\hat{\beta} = \arg\min_\beta \sum_t \left(y_t - \sum_i \beta_i x_{it}\right)^2 + \lambda \sum_i w_i \left(\frac{1}{2}|\beta_i| + \frac{1}{2}\beta_i^2\right)$, $w_i = 1/|\hat{\beta}_{i,\mathrm{OLS}}|$
  - $w_i$: adaptive weights, larger for coefficients small in OLS
  - The $|\beta_i|$ part sets coefficients to zero, the $\beta_i^2$ part shrinks them
  - Rolling 150-day windows: plain elastic net, $w_i = 1$, with $p = 1$ (the order BIC selects in 100% of the windows) and $H = 10$
- Why: selection and shrinkage in the VAR coefficients
  - without forcing a sparse **network** (links also come through $\Sigma$)
- On the full sample we keep the BIC order of our OLS VAR, $p = 2$
- With about 4 000 days the penalty hardly matters
  - it matters in short windows

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# Granger-Causality Networks and Multiple Testing (1/2)

- Billio, Getmansky, Lo & Pelizzon (2012) draw an edge $j \to i$ when $j$ Granger-causes $i$ at 5%
  - with 13 banks this is $13 \times 12 = 156$ tests
  - $j$ Granger-causes $i$ if the past of $j$ improves the forecast of $i$
  - In $X_{i,t} = c + aX_{i,t-\mathbf{1}} + gX_{j,t-\mathbf{1}} + u_t$ we test $H_\mathbf{0}$: $g = 0$
  - Even with no links at all, about 7.8 edges would appear by chance

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# Granger-Causality Networks and Multiple Testing (2/2)

- Our tests: bivariate regressions, one lag, heteroskedasticity-robust Wald statistics
  - Daily returns: 57 edges at 5%, 33 after the Holm (1979) correction, 55 with the false discovery rate controlled at 5% Benjamini & Hochberg (1995)
  - Log volatility, highly persistent: one extra lag that is not tested Toda & Yamamoto (1995), as in the lag-augmented tests of Shi, Hurn & Phillips (2020): 153, 150 and 153 edges
- Holm controls the chance of any false edge under arbitrary dependence
- The FDR is the **expected** share of false edges among those drawn
  - the BH (Benjamini–Hochberg) procedure guarantees it for independent tests Benjamini & Hochberg (1995)
  - but tests on overlapping bank pairs are dependent
- Daily returns of banks in different time zones
  - many "causal" edges only reflect that New York closes after Frankfurt

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# Worked Example: Reverse Stress Test with Two Factors

- Betas $b = (1.2, 0.8)$, target loss $\ell^* = 20\%$
  - four-week covariance of the factors $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ (%²)
  - $h\Sigma\, b = (1.2 \cdot 16 + 0.8 \cdot 6,\ 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
  - $b^\top h\Sigma\, b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, standard deviation 6.35%
- Most plausible scenario and its distance
  - $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%), \quad d^* = 20/6.35 = 3.15$
  - Normal probability of a loss at least this large: $\Phi(-3.15) = 0.08\%$ per four weeks
- The scenario loads more on the factor with the larger beta and variance
  - both factors fall together because they are correlated

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## Sectors, Factors and Lag Order

- Data (Section 3): the 17 financial sectors
  - A financial sector = equal-weighted CDS of its banks and insurers with assets of at least USD 10 billion, 199 institutions in all
- Factors (Eq. (8), Figure 2)
  - Residual pairs with $|\rho| < 0.25$: 91.6% with factors, fewer than half without
- Lag order and horizon (Sections 2.2 and 4)
  - VAR(1) chosen by BIC at the mean and kept at every quantile; $h = 5$ days
  - The quantile is held fixed along the forecast horizon

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## Further Results

- Of the 1 122 bilateral spillovers, 568 are below 1% at the mean and 829 at the median (Figure 7)
  - Away from the mean, the network has many weak links rather than a few strong ones
  - In the tails these weak links add up to strong average spillovers
- RTD over time (Section 4.3, Figure 11)
  - RTD rises with the Bear Stearns fund losses of 2007
  - It falls after the US bank rescue programme and the negative rates of the ECB: a fragility signal the mean hides

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## Replication Specification and Limits Stated by the Authors

- Specification for the replication on the 13 banks
  - VAR order by BIC, factors by Eq. (8), $h = 5$, $\tau \in \{0.05, 0.5, 0.95\}$ and the mean, $w = 250$, RTD
- Further limits stated by the authors
  - 250 days are too few to select factors window by window, so their number is fixed (Section 4.3, Appendix B of the paper)
  - Shorter windows leave too few effective tail observations
  - Only direct spillovers of idiosyncratic shocks are measured, not propagation through the common factors (footnote 2)
  - Greece is left out for a gap in its CDS data (Section 3, Appendix A of the paper)

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# The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: map what is known, then verify every reference
  - "Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls."
- **Hypotheses**: competing readings of a rise in connectedness, each with its own test
  - "Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them."

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# The Discovery Loop, Reusable Prompts (2/2)

- **Data and code**: on the course data, replicating a known number first
  - "Fit a VAR(2) to the log Parkinson volatilities of the 13 banks of the course data on 250-day windows and compute the generalized total connectedness with H = 10."
- **Robustness**, fixed before the predictive regressions are run
  - "Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018."
- **Critique**: ask the assistant to attack the result
  - "Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?"

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# What the Researcher Must Check (1/2)

- **Non-stationary inputs**: a VAR on price levels, or on trending series, produces spurious connectedness
  - Test: unit-root tests on every input
  - All eigenvalues of the companion matrix of the fitted VAR strictly inside the unit circle in every window
- **Ordering**: Cholesky decompositions change with the order of the banks
  - Test: permute the columns; the generalized table must not change

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# What the Researcher Must Check (2/2)

- **Look-ahead**: a window labelled by its first date, or a lag order chosen on the full sample
  - Test: the index at $t$ must not move when data after $t$ are deleted
- **Overlap**: rolling indices and forward outcomes overlap
  - so OLS $t$-statistics are inflated
  - Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension**: 13 banks and $p = 2$ give 27 coefficients per equation from 250 days
  - more banks need shrinkage Demirer et al. (2018)
  - Test: compare with a LASSO VAR
  - Unstable edges signal estimation noise, not structure

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# Further Study on Quantinar

## Advanced courses on quantinar.com

- Financial Risk Meter for Emerging Markets
- Dynamic Crypto Networks
- Network Centrality

## Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

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# Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **BAC**: Bank of America Corporation (NYSE ticker)
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BIC**: Bayesian Information Criterion
- **BNP**: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CATFIN**: CATastrophic risk in the FINancial sector
- **CDS**: Credit Default Swap
- **CI**: Confidence Interval
- **COVID-19**: COronaVIrus Disease 2019
- **CV**: Cross-Validation
- **DBK**: Deutsche Bank AG (Xetra ticker)
- **DCC**: Dynamic Conditional Correlation
- **EBA**: European Banking Authority
- **ECB**: European Central Bank
- **ES**: Expected Shortfall
- **EU**: European Union
- **FDIC**: Federal Deposit Insurance Corporation
- **FDR**: False Discovery Rate
- **FEVD**: Forecast Error Variance Decomposition
- **FRM**: Financial Risk Meter
- **GAAP**: Generally Accepted Accounting Principles (US)
- **GACV**: Generalized Approximate Cross-Validation
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GFC**: Global Financial Crisis (2007–2009)

# Acronyms used in this material (2/2)

- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GS**: The Goldman Sachs Group, Inc. (NYSE ticker)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HSBA**: HSBC Holdings plc (London Stock Exchange ticker)
- **HSBC**: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- **IFRS**: International Financial Reporting Standards
- **ING**: ING Groep (Internationale Nederlanden Groep) — *the Dutch banking group ING*
- **INGA**: ING Groep N.V. (Euronext Amsterdam ticker)
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **LRMES**: Long-Run Marginal Expected Shortfall
- **MES**: Marginal Expected Shortfall
- **MS**: Morgan Stanley (NYSE ticker)
- **OLS**: Ordinary Least Squares
- **QFVAR**: Quantile Factor Vector AutoRegression
- **QML**: Quasi-Maximum Likelihood
- **RTD**: Relative Tail Dependence
- **SAN**: Banco Santander SA (Bolsa de Madrid ticker)
- **SG**: Société Générale — *the French banking group Société Générale*
- **SRISK**: Systemic RISK (capital shortfall measure)
- **SVB**: Silicon Valley Bank
- **TENET**: Tail-Event driven NETwork risk
- **TLV**: Banca Transilvania (BVB ticker)
- **US**: United States
- **VAR**: Vector AutoRegression
- **VIX**: CBOE Volatility IndeX
- **WFC**: Wells Fargo & Company (NYSE ticker)

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